Pricing Additional Tier 1 Bonds as Claims on Bank Capital

A capital-ratio model with a stochastic refinancing spread, a Tier 2 anchor for the write-down hazard, and an application to BNP Paribas

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Abstract

Additional Tier 1 (AT1) bonds are perpetual, deeply subordinated bank bonds whose coupons, principal and redemption all depend on the issuer's capital. We model them as claims on the Common Equity Tier 1 (CET1) ratio, which follows a mean-reverting jump-diffusion, together with a correlated stochastic refinancing spread that drives both the issuer's call decision and a doubly-stochastic write-down hazard. Coupons are restricted by the Maximum Distributable Amount (MDA) rule along a time-varying regulatory threshold, principal converts at a contractual trigger, and the issuer calls only when refinancing is cheaper. Conditioning on the state paths makes the price an explicit, smooth, strictly decreasing and convex function of the hazard level, so the market-implied hazard is unique, is solved without simulation noise in the hazard direction, and every comparative static is computed with common random numbers. We then price the issuer's Tier 2 and senior non-preferred bonds on the same simulated state. Because the creditor hierarchy makes Tier 2 exposed to the same resolution event as AT1, this splits the AT1 hazard into a part that Tier 2 also prices and an AT1-specific premium. Calibrated to BNP Paribas's three USD AT1s on 25 September 2026, the bonds imply a hazard of 2.19% a year, equal across bonds to within the bid–offer; 1.72% is priced in BNP's Tier 2 and 0.47% is specific to the AT1. Of the $\approx$ 275bp spread, about 180bp is priced in Tier 2, 50bp is an AT1-specific premium (a lower bound) and 40–46bp are contract features: MDA coupon cuts, conversion and extension. The split is robust to a reputational call rule, to the order of attribution (Shapley values within blocks) and to a term structure of the resolution hazard. Prices are flat in capital above the MDA threshold and steep at it, so the risk-relevant distance is to the MDA threshold, not to the trigger. The speed of capital mean reversion is the weakest input and scales the capital stress results by a factor of five across its bootstrap interval.

Keywords: contingent convertibles, AT1, CoCos, bank capital, MDA, point of non-viability, Cox processes, conditional Monte Carlo, Shapley value.
JEL: G12, G13, G21, G28.

*Credit-Quant-Research, `at1-coco-pricer`. Code, tests and the scripts that produce every number and figure in this paper are in the repository. Market data are S&P Capital IQ evaluated prices and are not redistributed. Academic work, not investment advice.

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1 Introduction

Additional Tier 1 (AT1) bonds, often called contingent convertibles or CoCos, are the most equity-like instruments that European banks issue as debt. They count as going-concern Tier 1 capital under the Capital Requirements Regulation, they are perpetual and callable only by the issuer, their coupons are fully discretionary and non-cumulative, and their principal is converted into shares or written down when the bank's Common Equity Tier 1 (CET1) ratio falls below a contractual trigger. In addition, the resolution authority can impose losses on them at the point of non-viability (PONV), whatever the level of the trigger. The write-down of CHF 16 billion of Credit Suisse AT1s in March 2023, while the bank reported a CET1 ratio around 14%, made the last feature the dominant one in market pricing.

Every one of these cash-flow contingencies is written, directly or indirectly, on the bank's capital. The trigger is a CET1 level; the coupon restriction of the Maximum Distributable Amount (MDA) regime is triggered when CET1 falls inside the combined buffer requirement; regulatory approval of a call requires the bank to remain above its requirements after redemption; and the regulator's decision to declare the bank failing or likely to fail is, in practice, informed by the same deterioration of capital and market access. This paper takes that observation literally: it models the AT1 bond as a claim on the CET1 ratio, rather than as a straight bond with an equity barrier attached.

Why not an equity barrier. The workhorse market model for CoCos treats conversion as a barrier event on the share price (De Spiegeleer and Schoutens, 2012). That approach is convenient, because the share price is observed and its volatility is quoted, but it needs a mapping from the share price to the accounting ratio on which the contract is written. That mapping is not observable and not stable: banks' equity traded at a fraction of book value in 2023 with unchanged capital ratios, and at a premium in other periods. A model written on the CET1 ratio makes the trigger, the MDA restriction and the call constraint consistent by construction. Its cost is that CET1 is not traded and is observed quarterly with a lag, so its dynamics must be estimated under the physical measure and the residual risk premia must be absorbed elsewhere. We absorb them in a write-down hazard that is implied from bond prices, and we then ask how much of that hazard the bank's other bonds also price.

Capital alone is not enough. A model in which CET1 is the only state variable cannot price BNP Paribas's AT1s. With the historical volatility of BNP's CET1 ratio, the trigger is 7.8 percentage points away and is almost never reached in simulation: the bonds would be worth 114–118 against market prices of 94–98. The call decision, moreover, depends on the cost of refinancing in the market, which moves with sector spreads and not only with the bank's capital. We therefore add a second state variable: the issuer's new-issue AT1 spread, a mean-reverting Gaussian process correlated with capital shocks, with a capital add-on once CET1 enters the buffer. This spread does two jobs. It drives the call decision, so that the issuer extends exactly when refinancing is expensive, and it scales a doubly-stochastic write-down hazard, so that extension states are also the states in which the bond is riskier. Without the second link, extension would be valued with a discount rate that does not reflect the state, and the option not to call could even look valuable to the investor; we show this explicitly in Section 8.

What the hazard prices. The implied write-down hazard absorbs everything the mechanical features leave unexplained, and on its own it cannot tell the bank's credit risk apart from risk that is specific to the instrument. The bank's other bonds can. Under the EU resolution framework

the creditor hierarchy is respected: CET1 absorbs losses first, then AT1, then Tier 2. The joint statement of the Single Resolution Board, the EBA and ECB Banking Supervision of 20 March 2023 reaffirmed this ordering after the Credit Suisse case. Tier 2 bonds are therefore exposed to the same resolution event as AT1, at a better recovery, but have none of the AT1's coupon discretion, MDA restriction, trigger or perpetual extension. Pricing them on the same simulated state identifies how much of the AT1 hazard is compensation that Tier 2 also earns.

1.1 Contributions

The paper makes five contributions.

1. **A capital-ratio model with an endogenous call and a state-dependent hazard.** The CET1 ratio follows a mean-reverting jump-diffusion around the bank's target, the refinancing spread a correlated Ornstein–Uhlenbeck process, and the write-down time is a Cox process whose intensity is proportional to the effective spread. The MDA threshold is time-varying, which matters because scheduled buffer step-ups shrink the bank's cushion at constant capital.

2. **An exact conditional Monte Carlo for the hazard.** Conditioning on the state paths integrates the write-down time out analytically. The price becomes an explicit function $V(\lambda)$ that is smooth, strictly decreasing and strictly convex, so the implied hazard is unique and is found by a bracketing root-finder on a single set of paths. The same construction prices Tier 2 bonds with a recovery leg and a piecewise term structure of the hazard.

3. **A Tier 2 anchor for the AT1 hazard.** Writing the AT1 hazard as $\lambda = \lambda_{\text{res}} + \lambda_{\text{spec}}$, with two independent exponential clocks, gives the AT1 and the Tier 2 the right joint survival structure. Calibrating λ_{res} to the bank's Tier 2 and checking it against the senior non-preferred bond turns an unexplained premium into an observable part and an AT1-specific residual.

4. **An order-robust attribution of the spread.** Sequential decompositions of a credit spread into its components depend on the order in which features are switched on. We compute the exact Shapley value over all orders and a Shapley value within blocks, and show which conclusions survive. The order matters for the extension component and not for the split between Tier 2-priced and AT1-specific risk.

5. **An application with honest identification.** We calibrate the model to BNP Paribas's three USD AT1s, two Tier 2 bonds, one senior non-preferred bond and its senior CDS, with 50 quarters of CET1 history. We report Monte Carlo errors, price the identification limits of evaluated bid–offer quotes, bootstrap the capital dynamics and show which results depend on the weakly identified speed of capital mean reversion.

1.2 Main findings

On 25 September 2026 the three AT1s trade at yield-to-call spreads of 269–280bp over call-matched Treasuries. The model reprices them with hazards of 2.21%, 2.18% and 2.19% a year. The dispersion of 2.6bp of hazard is about four Monte Carlo standard errors but only 0.20–0.22 points of price, the size of the bid–offer: the bonds are consistent with a single hazard within the precision of the data, and cannot discriminate between model variants that differ by a few basis points.

BNP's Tier 2 2035NC34 implies a resolution hazard of 1.72% at the conventional 20% recovery, and

the senior non-preferred bond 1.63% at 40%: the two instruments that share the resolution event agree under the Markit recovery conventions, and the Tier 2 recovery that equalises them is 16%. The AT1 hazard therefore splits into 1.72% priced in Tier 2 and 0.47% specific to the AT1. In spread terms, about 180bp of the AT1 spread is priced in Tier 2, 50bp is an AT1-specific premium, and 40–46bp pays for the contract features: expected MDA coupon cuts ($\approx$ 30bp, an upper bound), mechanical conversion ($\approx$ 4bp) and extension (6–15bp, decreasing in the reset margin). Because the MDA component is an upper bound and the total is fixed by the market price, the AT1-specific premium is a lower bound.

The split survives three robustness checks. A reputational call rule raises the probability of a call at the first reset from 49–58% to 77–81%, close to the levels practitioners quote, but moves neither the hazard nor its split. Every attribution that prices the hazard before the contract features gives the same 181/50bp split. A two-piece resolution hazard bootstrapped from the two Tier 2 bonds (1.40% to 2030, 2.10% after) changes the Tier 2-priced component by less than 5bp.

Capital risk is a cliff, not a slope. At BNP's 13% CET1 ratio the price moves by about 0.6 points per point of capital; at the MDA threshold it moves by about four. Crossing the threshold switches on three mechanisms at once: coupons are cut, the call is blocked and the hazard rises. The distance that matters for risk management is the distance to the MDA threshold, which the scheduled O-SII step-ups reduce from 2.5 to 2.0 percentage points by 2028 at constant capital, not the distance to the trigger. The level of the capital stress depends strongly on the speed of capital mean reversion, which fifty quarters of history identify only to within a factor of eight; across that interval the price impact of the EBA adverse scenario ranges from -7 to -38 points.

1.3 Related literature

Contingent capital. The academic literature on contingent capital starts from structural models in which the trigger is written on the value of the bank's assets or on an accounting ratio. Pennacchi (2010) models bank assets as a jump-diffusion and prices CoCos with a trigger on the ratio of equity to assets; Glasserman and Nouri (2012) derive closed-form prices for contingent capital with a capital-ratio trigger and gradual conversion. Flannery (2014) surveys the design issues, including the choice between market and accounting triggers and the role of regulatory discretion. These models are built to study incentives and design, and typically do not include the MDA restriction, the issuer's call option or a regulatory write-down hazard independent of the trigger. Market practice instead relies on reduced-form or equity-barrier models (De Spiegeleer and Schoutens, 2012), in which the trigger is approximated by a barrier on the share price. Avdjiev et al. (2013) describe the market and the instrument features in detail. Our model sits between the two: the state is the accounting ratio on which the contract is written, as in the structural tradition, but the regulatory and market features that dominate AT1 pricing today (discretionary coupons, MDA, issuer calls, PONV) are modelled explicitly and the residual risk is priced in reduced form.

Reduced-form credit risk. The write-down time is a doubly-stochastic (Cox) process in the sense of Lando (1998), with intensity driven by the state, as in the reduced-form models of Jarrow and Turnbull (1995) and Duffie and Singleton (1999); Bielecki and Rutkowski (2002) give the general construction. Our use of the construction is specific in two respects. First, the intensity is proportional to the issuer's simulated refinancing spread, which also drives the call, so that the hazard and the extension decision are consistent state by state. Second, we split the intensity into two independent clocks to price the AT1 and the Tier 2 jointly, which is the reduced-form counterpart of the creditor hierarchy.

Monte Carlo methods. Integrating the default time out conditionally on the state paths is a form of conditional Monte Carlo, or Rao–Blackwellisation (Glasserman, 2004); it removes the variance due to the default event and makes the price smooth in the hazard. Discrete monitoring of the trigger introduces the usual bias of barrier options, which Broadie et al. (1997) correct with a shift of the barrier; we show that the correction is negligible here.

Decomposition and attribution. Attributing a spread to its components by switching features on one at a time is order-dependent whenever the features interact. The Shapley value (Shapley, 1953) resolves the dependence by averaging over all orders, and Shorrocks (2013) applies it to decompositions in economics; the value for games with a priori unions (Owen, 1977) handles groups of related factors. We use both ideas and show that, for AT1 bonds, the economically consistent attribution prices the hazard before the contract features.

1.4 Outline

Section 2 describes the instrument, the regulatory framework and the three BNP Paribas bonds. Section 3 sets out the model and derives the properties of the state processes. Section 4 derives the pricing equations, the conditional Monte Carlo estimator and its properties, and the pricing of Tier 2 and senior bonds. Section 5 defines the spread attribution. Section 6 describes the data and the calibration. Section 7 reports the results, Section 8 the robustness checks and Section 9 discusses implications and limitations. The appendices contain proofs, numerical diagnostics and the parameter table.

2 The instrument and the regulatory framework

This section describes the features of AT1 bonds that the model must reproduce, the regulatory rules behind them, and the three BNP Paribas bonds studied in the paper. References to articles are to the Capital Requirements Regulation (CRR, Regulation (EU) No 575/2013), the Capital Requirements Directive (CRD, Directive 2013/36/EU) and the Bank Recovery and Resolution Directive (BRRD, Directive 2014/59/EU), as amended.

2.1 Place in the capital structure

A bank's own funds are layered. Common Equity Tier 1 (CET1) is ordinary share capital and reserves; Additional Tier 1 (AT1) instruments complete Tier 1 capital, the going-concern layer; Tier 2 instruments are gone-concern capital, subordinated debt with an original maturity of at least five years. Below own funds, the bank's loss-absorbing capacity in resolution (TLAC for global systemically important banks, MREL for all banks) is completed by senior non-preferred debt, which ranks ahead of Tier 2 but behind senior preferred debt and deposits.

In a liquidation or a resolution the creditor hierarchy determines the order in which losses are absorbed: CET1 first, then AT1, then Tier 2, then senior non-preferred, then senior preferred. For pricing purposes this ordering has a precise implication that we use in Section 4.4: every resolution event that imposes losses on Tier 2 has already wiped out the AT1, while the converse is not true. AT1 holders are exposed to all resolution events and, in addition, to events that hit AT1 alone.

2.2 Eligibility and the perpetual call structure

To count as AT1, an instrument must be perpetual, with no incentive to redeem (CRR Art. 52). The issuer may call it no earlier than five years after issuance, and only with the prior permission of the competent authority (Art. 77). Permission requires that the instrument be replaced by own funds of equal or higher quality on terms that are sustainable for the bank's income, or that the bank demonstrate that its own funds would exceed its requirements by a margin the authority considers necessary (Art. 78).

The bonds studied here are callable only on their reset dates, every five years from the first call. If not called, the coupon resets to the prevailing five-year US Treasury constant-maturity yield plus the margin fixed at issuance. The fixed margin is the source of extension risk: the issuer calls when it can refinance at a lower spread, and does not call when spreads have widened, which is when investors would most like their money back. Because the margin is fixed and the reference rate floats, the economic call decision compares the reset margin with the issuer's new-issue AT1 spread over the same benchmark; the level of rates cancels.

In practice, European issuers have called AT1s at the first opportunity in almost all cases, partly for reputational reasons: a bank that does not call signals that its funding is expensive. The best-known exception, Banco Santander's decision in early 2019 not to call a euro AT1, was widely reported precisely because it was rare. The model's base call rule is purely economic; Section 8 adds a reputational tolerance.

2.3 Discretionary coupons and the Maximum Distributable Amount

AT1 coupons are fully discretionary and non-cumulative: the issuer may cancel any coupon at any time, cancelled coupons are lost, and cancellation is not an event of default (CRR Art. 52(1)(l)). Coupons can only be paid out of distributable items.

In addition, coupons are mechanically restricted when the bank breaches its combined buffer requirement (CRD Art. 141). Let the CET1 requirement be the sum of the Pillar 1 minimum (4.5% of risk-weighted assets), the CET1 part of the Pillar 2 requirement and the combined buffer requirement (capital conservation, countercyclical, systemic risk and G-SII/O-SII buffers). A bank whose CET1 ratio falls below this total may distribute (dividends, variable remuneration and AT1 coupons together) at most the Maximum Distributable Amount, a fraction of its interim and year-end profits that depends on where the ratio sits inside the combined buffer:

Position of CET1 inside the combined buffer	Maximum distributable fraction of profits
top quartile	0.6
second quartile	0.4
third quartile	0.2
bottom quartile	0

Two features of the rule matter for modelling. First, the MDA caps *all* distributions together; a bank inside the buffer chooses how to allocate the amount, and would normally cut dividends, buy-backs and variable pay before cutting AT1 coupons, which are small relative to profits and whose cancellation is costly for market access. Applying the quartile factor to the coupon itself, as our model does, therefore overstates the expected coupon loss. Second, analogous restrictions apply when the bank breaches its leverage-ratio buffer (L-MDA) or its MREL requirement (M-MDA); for some banks these bind before the CET1-based MDA.

2.4 Loss absorption: trigger and point of non-viability

Each AT1 contains a mechanical trigger: if the issuer's CET1 ratio falls below a contractual level (at least 5.125%, CRR Art. 54), the instrument is converted into ordinary shares or written down, permanently or temporarily. The bonds studied here convert into shares at the higher of the prevailing market price and a floor price set at issuance, so that the conversion ratio is capped.

Independently of the trigger, the resolution authority must write down or convert capital instruments when the bank reaches the point of non-viability (BRRD Art. 59): when the institution is failing or likely to fail and no alternative measure would prevent its failure, or when public support is required. The write-down follows the creditor hierarchy (BRRD Art. 48): CET1 first, then AT1, then Tier 2. A PONV write-down can therefore occur while the CET1 ratio is far above the trigger. The Credit Suisse case of March 2023, decided under Swiss law, wrote AT1 holders down to zero while shareholders retained value; the EU authorities' joint statement of 20 March 2023 stated that in the EU, losses would continue to be absorbed by CET1 first and AT1 only after CET1 had been fully used.

For pricing, this means two things. First, the PONV event is the dominant risk for a well-capitalised issuer: the trigger is far away, but the regulator's decision is not tied to it. Second, in the EU the PONV event that hits AT1 is, in most scenarios, the same event that hits Tier 2. AT1 holders lose everything, Tier 2 holders lose part of their claim. The AT1-specific part of the risk is the probability that AT1 is written down while Tier 2 is not, together with the risk premia that investors attach to the equity-like features of the instrument.

2.5 The BNP Paribas bonds

BNP Paribas is a global systemically important bank headquartered in France, supervised by the ECB and resolvable by the Single Resolution Board. Its CET1 ratio was 12.97% at 30 June 2026 against a published CET1 requirement of 10.43%; the 2025 requirement stack consists of Pillar 1 (4.5%), the CET1 part of Pillar 2R (1.14%), the capital conservation buffer (2.5%), the G-SIB buffer (1.5%), the countercyclical buffer (0.73%) and the systemic-risk buffer (0.14%). Under the French O-SII framework the bank's O-SII buffer rises by 25bp on 1 January 2027 and by a further 25bp on 1 January 2028, and the higher of the G-SIB and O-SII buffers applies; at constant capital these step-ups reduce the cushion above the MDA threshold from 2.5 to 2.0 percentage points.

Table 1 lists the three USD AT1s and the three bonds used as the Tier 2 and senior anchor. The AT1s share the issuer, the trigger (5.125% of group CET1), the loss-absorption mechanism (conversion into ordinary shares) and the reset reference (five-year US Treasury CMT); they differ in coupon, first call date and reset margin. The Tier 2 and senior non-preferred (SNP) notes are callable once, one year before maturity or at a reset date, when they cease to count as Tier 2 or MREL; the market prices these calls as certain. The Tier 2 2035NC34 is the anchor: its call in November 2034 falls inside the AT1 call window (December 2033 to April 2036).

Bond	ISIN (144A)	Rank	Coupon	Call	Maturity	Reset margin
6.875% NC33	US05602XQR25	AT1	6.875%	15-Dec-2033	perpetual	UST5 + 285bp
7.20% NC36	US05602XQS08	AT1	7.200%	17-Apr-2036	perpetual	UST5 + 294bp
7.45% NC35	US05602XQQ42	AT1	7.450%	27-Jun-2035	perpetual	UST5 + 313bp
T2 2035NC34	US09659T2D24	T2	5.906%	19-Nov-2034	19-Nov-2035	SOFR + 192bp
T2 2035NC30	US09659T2B67	T2	2.588%	12-Aug-2030	12-Aug-2035	UST5 + 205bp
SNP 2035NC34	US09659W2X16	SNP	5.738%	20-Feb-2034	20-Feb-2035	SOFR + 188bp

Table 1: Bonds used in the paper. AT1s: perpetual, callable on reset dates every five years, trigger 5.125% of group CET1, conversion into ordinary shares at the higher of the market price and a floor (USD 61.12, 72.52 and 61.17 per share respectively). Tier 2 and SNP: single call, treated as certain. Sources: BNP Paribas term sheets and debt issuance list.

Figure 1 shows the AT1 prices over the sample. The 7.45% has traded since June 2025, the 6.875% since December 2025 and the 7.20% since April 2026. All three fell by six to seven points over 2026, most of it from rising Treasury yields; in the four weeks to 25 September 2026 the ten-year Treasury rose 45bp and the spread of the low-margin 6.875% widened by a further 34bp, against 10–12bp for the other two.

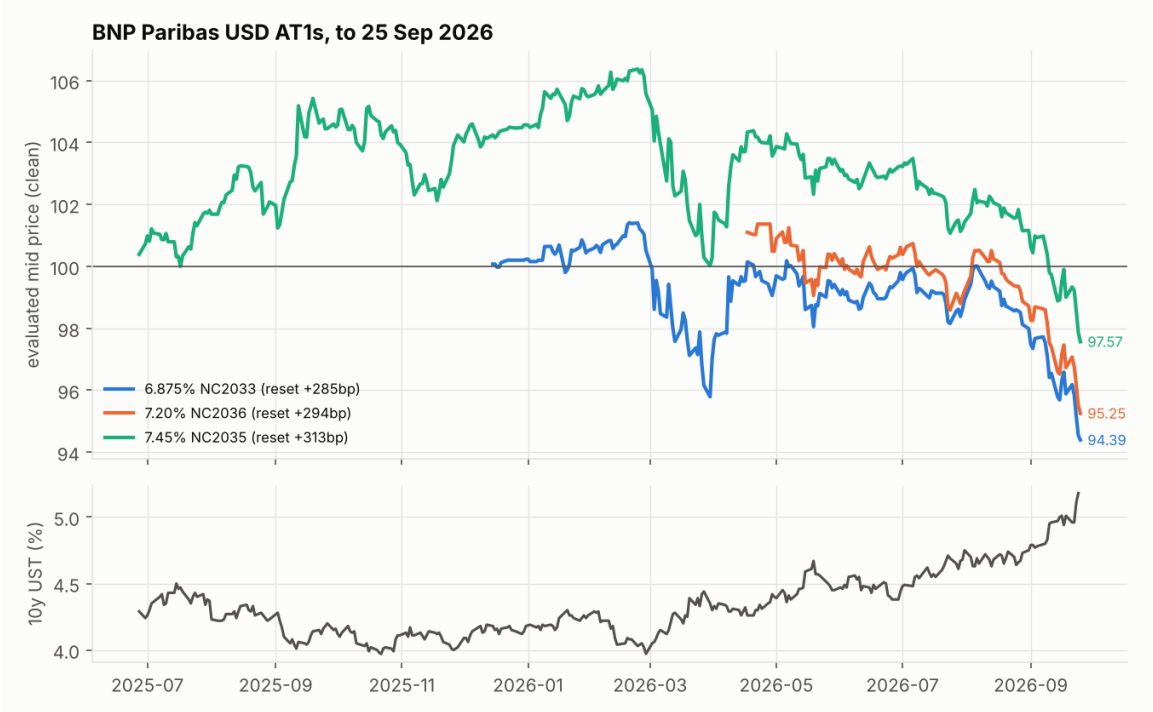


Figure 1: BNP Paribas USD AT1s: Capital IQ evaluated mid prices.

3 The model

3.1 Probability space and measures

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ satisfying the usual conditions, supporting two standard Brownian motions W and B with $\mathrm{d}\langle W, B\rangle_t = \rho\,\mathrm{d}t$, a Poisson process N with intensity

λ_J, an i.i.d. sequence of jump sizes $(J_n)_{n\geq 1}$, and two unit exponential random variables E_{res} and E_{spec}. The Brownian motions, the compound Poisson process and the exponentials are mutually independent, except for the correlation between W and B. Denote by $\mathcal{G} = (\mathcal{G}_t)$ the filtration generated by $(W, B, N, (J_n))$, the information in the state paths, and by $\mathcal{G}_\infty = \bigvee_t \mathcal{G}_t$ its terminal σ-algebra.

All prices are expectations under $\mathbb{Q}$, a pricing measure. The CET1 ratio is not a traded asset and its dynamics are estimated from quarterly history, that is under the physical measure $\mathbb{P}$.

Assumption 3.1 (Capital dynamics under the pricing measure)**.** The dynamics of the CET1 ratio are the same under $\mathbb{Q}$ and under $\mathbb{P}$. All risk premia not captured by the mechanical features of the contract are absorbed by the write-down hazard λ, which is therefore a risk-neutral intensity and not a probability of default.

Assumption 3.1 is the standard compromise of reduced-form models calibrated to an untraded state variable. Its consequence is that λ should be read as a price, not as a forecast: a hazard of 2.2% a year is the compensation the market requires for the write-down risk and its premium, not a 2.2% annual probability of write-down.

3.2 The CET1 ratio

Let C_t be the CET1 ratio in percentage points. It follows the mean-reverting jump-diffusion

$$\mathrm{d}C_t = \kappa(\theta - C_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t - \mathrm{d}L_t, \qquad L_t = \sum_{n=1}^{N_t} J_n, \tag{1}$$

with $\kappa > 0$, $\sigma > 0$, and lognormal jump sizes with mean m_J,

$$J_n = m_J \exp\!\big(\sigma_J Z_n - \tfrac{1}{2}\sigma_J^2\big), \qquad Z_n \sim N(0,1), \qquad \mathbb{E}[J_n] = m_J, \quad \mathbb{E}[J_n^2] = m_J^2 e^{\sigma_J^2}. \tag{2}$$

The drift represents capital management: retained earnings, distribution policy and the management of risk-weighted assets pull the ratio back to the target θ, the CET1 objective the bank communicates to investors. Jumps are one-sided because losses (a fine, a trading loss, an impairment, a regulatory change in risk weights) arrive suddenly, while capital is rebuilt gradually.

Proposition 3.2 (Solution and moments)**.** *Equation* (1) *has the unique strong solution*

$$C_t = \theta + (C_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u - \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}L_u, \tag{3}$$

with mean and variance

$$\mathbb{E}[C_t] = \theta + (C_0 - \theta)e^{-\kappa t} - \frac{\lambda_J m_J}{\kappa}\big(1 - e^{-\kappa t}\big), \tag{4}$$

$$\mathrm{Var}[C_t] = \frac{\sigma^2 + \lambda_J m_J^2 e^{\sigma_J^2}}{2\kappa}\big(1 - e^{-2\kappa t}\big). \tag{5}$$

As $t \to \infty$, C_t converges in distribution to a stationary law with mean $\theta - \lambda_J m_J/\kappa$ and variance $(\sigma^2 + \lambda_J m_J^2 e^{\sigma_J^2})/(2\kappa)$.

Proof. The coefficients are affine with Lipschitz drift and constant diffusion, so a unique strong solution exists. Apply Itô's formula for semimartingales to $e^{\kappa t}C_t$: $\mathrm{d}(e^{\kappa t}C_t) = e^{\kappa t}(\kappa\theta\,\mathrm{d}t + \sigma\,\mathrm{d}W_t - \mathrm{d}L_t)$, and integrate to obtain (3). For the mean, the stochastic integral against W has zero expectation, and $\mathbb{E}\int_0^t e^{-\kappa(t-u)}\,\mathrm{d}L_u = \lambda_J m_J \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}u$ by the compensator of the compound Poisson

process. For the variance, the Brownian and jump integrals are independent; Itô's isometry gives $\sigma^2 \int_0^t e^{-2\kappa(t-u)}\,\mathrm{d}u$ for the first, and the isometry for compensated compound Poisson integrals gives $\lambda_J \mathbb{E}[J^2] \int_0^t e^{-2\kappa(t-u)}\,\mathrm{d}u$ for the second. Evaluating the integrals gives (4)–(5). The characteristic function of (3) converges as $t \to \infty$, which gives the stationary law. □

Remark 3.3 (Target and long-run mean). The long-run mean is not θ but $\theta - \lambda_J m_J/\kappa$: jumps pull the ratio down on average, and mean reversion offsets them only partially. With the base-case parameters ($\theta = 13\%$, $\kappa = 0.35$, $\lambda_J = 0.12$, $m_J = 2.2$pp) the stationary mean is 12.25%, and the stationary standard deviation is 1.17pp, of which the diffusion alone contributes $\sigma/\sqrt{2\kappa} = 0.56$pp. Interpreting θ as the bank's target is therefore accurate in the absence of shocks; in the presence of shocks, the model's capital hovers about 0.75pp below target on average. A calibration that wanted the long-run mean, rather than the shock-free target, to equal 13% would set $\theta = 13.75\%$; we keep θ at the communicated target and report the implication.

The mechanical trigger time is the first passage

$$\tau_m = \inf\{t \geq 0 :\ C_t \leq c_{\text{trig}}\}, \qquad c_{\text{trig}} = 5.125\%. \tag{6}$$

3.3 The refinancing spread

Let S_t be the market component of the issuer's new-issue AT1 spread, in decimal. It follows the Ornstein–Uhlenbeck process

$$\mathrm{d}S_t = a(\bar{s} - S_t)\,\mathrm{d}t + \eta\,\mathrm{d}B_t, \tag{7}$$

correlated with the diffusive capital shocks through $\rho < 0$: when capital falls, spreads tend to widen. The issuer's effective spread adds a capital premium once the CET1 ratio enters the buffer,

$$s_t = S_t + \beta\big(\bar{c}(t) - C_t\big)^+, \tag{8}$$

where $\bar{c}(t)$ is the MDA threshold defined in (11) below and β is the extra spread per point of CET1 inside the buffer. The effective spread is the rate at which the bank could issue a new AT1 at time t.

Proposition 3.4 (Spread distribution)**.** *The solution of* (7) *is Gaussian,*

$$S_t = \bar{s} + (S_0 - \bar{s})e^{-at} + \eta \int_0^t e^{-a(t-u)}\,\mathrm{d}B_u \ \sim\ N\Big(\bar{s} + (S_0 - \bar{s})e^{-at},\ \frac{\eta^2}{2a}\big(1 - e^{-2at}\big)\Big), \tag{9}$$

and its covariance with the diffusive part of capital, $C_t^W = \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u$*, is*

$$\mathrm{Cov}\big(S_t, C_t^W\big) = \frac{\rho\sigma\eta}{a + \kappa}\Big(1 - e^{-(a+\kappa)t}\Big). \tag{10}$$

In the stationary regime $\mathbb{Q}(S_t < 0) = \Phi\big(-\bar{s}\sqrt{2a}/\eta\big)$.

Proof. The solution follows as in Proposition 3.2. The covariance follows from Itô's isometry for two Wiener integrals: $\mathbb{E}[\int_0^t f\,\mathrm{d}B \int_0^t g\,\mathrm{d}W] = \rho \int_0^t f(u)g(u)\,\mathrm{d}u$ with $f(u) = \eta e^{-a(t-u)}$ and $g(u) = \sigma e^{-\kappa(t-u)}$. □

With the base-case parameters ($a = 0.5$, $\bar{s} = 275$bp, $\eta = 100$bp) the stationary standard deviation of S is 100bp and the stationary probability of a negative spread is 0.3%. Because the hazard and the call rule use the spread, negative values are handled explicitly: the hazard is floored at zero (equation (15)), and a negative refinancing spread simply triggers a call. A square-root or log-normal spread would avoid negative values at the cost of the exact Gaussian transition; at a probability of 0.3% the difference is immaterial.

3.4 The regulatory threshold

Let b be the MDA floor (Pillar 1 plus the CET1 part of Pillar 2R) and $w(t)$ the combined buffer at time t. The MDA threshold is

$$\bar{c}(t) = b + w(t), \qquad w(t) = w_0 + \sum_k \Delta w_k \, \mathbf{1}\{t \geq t_k\}, \tag{11}$$

where the steps $(t_k, \Delta w_k)$ are scheduled buffer changes. For BNP, $b = 5.64\%$, $w_0 = 4.79\%$ (so that $\bar{c}(0)$ equals the published requirement of 10.43%), and the O-SII step-ups give $\Delta w = 0.25$pp on 1 January 2027 and on 1 January 2028.

3.5 Contract features as functionals of the state

MDA coupon restriction. With $q_t = (C_t - b)/w(t)$ the position of the ratio inside the buffer, the fraction of the scheduled coupon that can be paid is

$$\phi(C_t, t) = \begin{cases} 1 & C_t \geq \bar{c}(t), \\ 0.6 & q_t \in [0.75, 1), \\ 0.4 & q_t \in [0.50, 0.75), \\ 0.2 & q_t \in [0.25, 0.50), \\ 0 & q_t < 0.25. \end{cases} \tag{12}$$

This applies the MDA quartile factors to the coupon itself. It is a stylised and conservative version of the rule, which caps all distributions together (Section 2); the resulting MDA component of the spread is an upper bound.

Call and extension. Let $T_1 < T_2 < \dots$ be the reset dates and m the reset margin. The issuer calls at

$$\tau_c = \min\{T_k : \; m > s_{T_k} - \delta, \;\; C_{T_k} \geq \bar{c}(T_k), \;\; T_k < \tau_m\} \qquad (\min \emptyset = +\infty), \tag{13}$$

that is, at the first reset date at which keeping the bond costs more than refinancing it, less a reputational tolerance $\delta \geq 0$, provided that the regulator would approve the call (CET1 above the threshold) and the bond is still alive. In the base case $\delta = 0$. Since each condition is $\mathcal{G}_{T_k}$-measurable, τ_c is a $\mathcal{G}$-stopping time taking values in $\{T_k\} \cup \{+\infty\}$. If $\eta > 0$ the call is random even for a healthy bank, and the bond with the lowest margin is the first to be extended when spreads widen.

Conversion. At τ_m the bond converts into shares at the higher of the market price and the floor price P_{floor}. When the share trades below the floor, as it will at the trigger, the investor receives $F \cdot S^{\text{eq}}_{\tau_m}/P_{\text{floor}}$. We do not model the share price jointly with capital; instead the share price at conversion is a fixed fraction ξ of today's price S^{eq}_0 converted at today's exchange rate X_0, so the recovery at conversion is

$$R_{\text{conv}} = \min\left(1, \; \frac{\xi \, S^{\text{eq}}_0 X_0}{P_{\text{floor}}}\right). \tag{14}$$

With $\xi = 0.2$, a share price of EUR 98.43 and $X_0 = 1.15$, R_{conv} is 37% for the 6.875% and the 7.45% and 31% for the 7.20%, whose floor is higher.

3.6 The write-down hazard

The regulator may impose losses before the trigger. We model the PONV time as a doubly-stochastic (Cox) time with intensity proportional to the effective spread,

$$\lambda_t = \lambda \, \frac{s_t^+}{s_0}, \qquad I_t = \int_0^t \frac{s_u^+}{s_0} \, \mathrm{d}u, \tag{15}$$

so that λ is today's hazard and I_t is the integrated driver. The hazard rises when spreads widen or capital falls inside the buffer. The recovery of the AT1 at PONV is zero.

Linking the hazard to s_t is what makes extension consistent. The issuer extends when $s_t > m$; with a state-dependent hazard those are exactly the states in which the bond is riskier, and the extended cash flows are discounted accordingly. With a constant hazard, the extended coupon $y^{\mathrm{ref}} + m$ would be discounted at a rate that does not reflect the state, and in states where m exceeds the risk-adjusted discount spread, not calling would add value for the investor. Section 8 shows the size of this effect.

3.7 Two clocks: resolution and AT1-specific write-down

To price the AT1 jointly with the bank's Tier 2, decompose the hazard level into

$$\lambda = \lambda_{\mathrm{res}} + \lambda_{\mathrm{spec}}, \qquad \lambda_{\mathrm{res}}, \lambda_{\mathrm{spec}} \geq 0, \tag{16}$$

and define two write-down times with the same driver,

$$\tau_{\mathrm{res}} = \inf\{t : \ \lambda_{\mathrm{res}} I_t \geq E_{\mathrm{res}}\}, \qquad \tau_{\mathrm{spec}} = \inf\{t : \ \lambda_{\mathrm{spec}} I_t \geq E_{\mathrm{spec}}\}. \tag{17}$$

The resolution time τ_{res} imposes losses on AT1 and on Tier 2; the AT1-specific time τ_{spec} imposes losses on AT1 alone. The AT1 write-down time is $\tau_P = \tau_{\mathrm{res}} \wedge \tau_{\mathrm{spec}}$.

Proposition 3.5 (Joint survival)**.** *Conditionally on $\mathcal{G}_\infty$,*

$$\mathbb{Q}(\tau_P > t \mid \mathcal{G}_\infty) = e^{-\lambda I_t}, \tag{18}$$

$$\mathbb{Q}(\tau_{\mathrm{res}} > t \mid \mathcal{G}_\infty) = e^{-\lambda_{\mathrm{res}} I_t}, \tag{19}$$

$$\mathbb{Q}(\tau_{\mathrm{res}} \leq t, \ \tau_{\mathrm{res}} \leq \tau_{\mathrm{spec}} \mid \mathcal{G}_\infty) = \frac{\lambda_{\mathrm{res}}}{\lambda}\big(1 - e^{-\lambda I_t}\big). \tag{20}$$

In particular τ_P is a Cox time with intensity $\lambda s_t^+/s_0$, and the probability that an AT1 write-down is a resolution event that also hits Tier 2 is $\lambda_{\mathrm{res}}/\lambda$, independently of the path.

Proof. I_t is $\mathcal{G}_\infty$-measurable, continuous and non-decreasing, and the exponentials are independent of $\mathcal{G}_\infty$ and of each other. Hence $\mathbb{Q}(\tau_{\mathrm{res}} > t \mid \mathcal{G}_\infty) = \mathbb{Q}(E_{\mathrm{res}} > \lambda_{\mathrm{res}} I_t \mid \mathcal{G}_\infty) = e^{-\lambda_{\mathrm{res}} I_t}$, which is (19); the same holds for τ_{spec} with λ_{spec}, and by conditional independence $\mathbb{Q}(\tau_P > t \mid \mathcal{G}_\infty) = e^{-\lambda_{\mathrm{res}} I_t} e^{-\lambda_{\mathrm{spec}} I_t}$, which is (18). For (20), change time to $u = I_t$: conditionally on $\mathcal{G}_\infty$, τ_{res} and τ_{spec} become independent exponential times in u with rates λ_{res} and λ_{spec}, and the probability that the first of two independent exponentials occurs before u and is the one with rate λ_{res} is $\frac{\lambda_{\mathrm{res}}}{\lambda}(1 - e^{-\lambda u})$. □

Proposition 3.5 is the formal content of the Tier 2 anchor. The AT1 price depends on λ_{res} and λ_{spec} only through their sum, so the AT1 alone cannot separate them; the Tier 2 depends on λ_{res} alone and identifies it. The ratio $\lambda_{\mathrm{res}}/\lambda$ is the share of AT1 write-down risk that the market prices in Tier 2 as well.

Two remarks on interpretation. First, λ_{res} is calibrated to the Tier 2 spread, which also contains the risk and liquidity premia that investors require on Tier 2 itself. "Priced in Tier 2" is therefore the accurate label for λ_{res}; "bank resolution risk" is an approximation. Second, the construction assumes that resolution hits AT1 whenever it hits Tier 2, which is the creditor hierarchy, and allows AT1-only write-downs of the Credit Suisse type through τ_{spec}. It rules out a Tier 2 loss without an AT1 loss, which the hierarchy also rules out.

4 Valuation

4.1 The pricing equation

Let r be the flat, continuously compounded risk-free rate used for discounting, F the face value, f the coupon frequency and $t_1 < t_2 < \dots$ the coupon dates. The scheduled coupon rate is $c(t_i) = c_0$ up to and including the first reset T_1 and $c(t_i) = y^{\text{ref}} + m$ after it, where y^{ref} is the five-year reference rate. A coupon is paid if the bond has not been converted, called or written down, scaled by the MDA fraction. The principal is repaid at the call, converted at the trigger, or lost at PONV. The dirty price at time 0 is

$$\begin{aligned} V_0 = \mathbb{E}\Bigg[& \sum_{t_i \le H} \mathbf{1}_{\{t_i < \tau_m,\ t_i \le \tau_c,\ t_i < \tau_P\}}\, \phi(C_{t_i}, t_i)\, \frac{c(t_i) F}{f}\, e^{-r t_i} + \mathbf{1}_{\{\tau_c < \tau_m \wedge \tau_P,\ \tau_c \le H\}}\, F\, e^{-r\tau_c} \\ & + \mathbf{1}_{\{\tau_m < \tau_c \wedge \tau_P,\ \tau_m \le H\}}\, R_{\text{conv}} F\, e^{-r\tau_m} + \mathbf{1}_{\{\tau_c \wedge \tau_m \wedge \tau_P > H\}}\, T_H\, e^{-rH} \Bigg], \end{aligned} \tag{21}$$

where H is the simulation horizon and T_H the value at H of a bond still alive,

$$T_H = \frac{(y^{\text{ref}} + m) F}{r + \lambda\, s_H^+ / s_0}. \tag{22}$$

The tail is the value of a perpetuity paying the reset coupon continuously, discounted at the risk-free rate plus the hazard frozen at its level at H. With $H = 40$ years and $r \approx 5\%$, the tail is discounted by $e^{-rH} \approx 0.13$ and weighted by the probability of survival without call to year 40, which is below 3% in the base case (Table 7); its contribution to the price is a few hundredths of a point.

4.2 Conditional Monte Carlo

The write-down time enters (21) only through the indicators $\mathbf{1}_{\{t < \tau_P\}}$. Every other quantity is a functional of the state paths.

Theorem 4.1 (Conditional pricing formula)**.** *Let* CF_i *be the coupon cash flows of* (21) *with the indicator of* τ_P *removed, and let* CF_{end} *and* T_{end} *be the redemption, conversion or tail payment and its date on each path. Then*

$$V_0(\lambda) = \mathbb{E}\left[\sum_i \text{CF}_i\, e^{-r t_i - \lambda I_{t_i}} + \text{CF}_{\text{end}}\, e^{-r T_{\text{end}} - \lambda I_{T_{\text{end}}}} \right]. \tag{23}$$

Proof. All of CF_i, CF_{end}, T_{end}, τ_m and τ_c are $\mathcal{G}_\infty$-measurable. By the tower property, $\mathbb{E}[X \mathbf{1}_{\{t < \tau_P\}}] = \mathbb{E}[X\, \mathbb{Q}(\tau_P > t \mid \mathcal{G}_\infty)] = \mathbb{E}[X e^{-\lambda I_t}]$ for any $\mathcal{G}_\infty$-measurable integrable X, by (18). Apply this to each term of (21). In the tail term the event $\{\tau_P > H\}$ contributes $e^{-\lambda I_H}$, and the value of the perpetuity after H is given by (22). □

The estimator replaces the simulated indicator by its conditional expectation. Two consequences follow.

Proposition 4.2 (Variance reduction)**.** *Let X be the discounted payoff of* (21) *on one path, with τ_P simulated, and $Y = \mathbb{E}[X \mid \mathcal{G}_\infty]$ the integrand of* (23)*. Then $\mathbb{E}[Y] = \mathbb{E}[X]$ and $\mathrm{Var}(Y) = \mathrm{Var}(X) - \mathbb{E}[\mathrm{Var}(X \mid \mathcal{G}_\infty)] \le \mathrm{Var}(X)$, with equality only if X is $\mathcal{G}_\infty$-measurable.*

Proof. This is the law of total variance. □

Proposition 4.3 (Shape of the price in the hazard)**.** *On any fixed set of paths, and in expectation, the map $\lambda \mapsto V_0(\lambda)$ on $[0, \infty)$ is infinitely differentiable, strictly decreasing and strictly convex, with*

$$V_0'(\lambda) = -\mathbb{E}\Big[\sum_i \mathrm{CF}_i \, I_{t_i} e^{-rt_i - \lambda I_{t_i}} + \mathrm{CF}_{\mathrm{end}} \, I_{T_{\mathrm{end}}} e^{-rT_{\mathrm{end}} - \lambda I_{T_{\mathrm{end}}}}\Big] + \textit{(tail term)} < 0, \tag{24}$$

$V_0(\lambda) \to 0$ as $\lambda \to \infty$. Consequently, for any market price $P \in (0, V_0(0))$ the implied hazard $\lambda^\star$ solving $V_0(\lambda^\star) = P$ exists and is unique, and any bracketing method on $[0, \bar\lambda]$ with $V_0(\bar\lambda) < P$ converges to it.

Proof. Each summand has the form $a\, e^{-\lambda b}$ with $a \ge 0$ and $b = I_t > 0$ for $t > 0$. The driver starts at $s_0/s_0 = 1$ (when CET1 is above the threshold) and is continuous, so $I_t > 0$ for every $t > 0$ on every path. Each summand is therefore smooth, decreasing and convex in λ, strictly so when $a > 0$. The tail summand is the product of $(y^{\mathrm{ref}} + m)F/(r + \lambda d)$ and $e^{-\lambda I_H}$ with $d = s_H^+/s_0 \ge 0$; both factors are positive, decreasing and convex, and for two such functions $(gh)'' = g''h + 2g'h' + gh'' \ge 0$ since $g'h' \ge 0$. Sums and averages preserve these properties, and at least one coupon is positive with positive probability, which gives strictness. Every summand vanishes as $\lambda \to \infty$; dominated convergence gives the limit. Existence and uniqueness of the root follow from continuity and strict monotonicity. □

In practice we solve $V_0(\lambda) = P^{\mathrm{clean}} + \mathrm{AI}$ with Brent's method (Brent, 1973) on $[0, 1]$. Because the path-wise cash flows and the integrated drivers are stored once, each evaluation of $V_0(\lambda)$ is a vectorised sum, and the root costs a few milliseconds. Proposition 4.3 also gives the price sensitivity to the hazard, which we use in Section 6 to translate hazard differences into price units: for the three AT1s $|V_0'|$ is 0.077–0.084 points per basis point of hazard.

4.3 Monte Carlo error of the implied hazard

Let $\hat V_n(\lambda)$ be the sample mean of n paths. By the delta method, the standard error of the implied hazard is approximately

$$\mathrm{se}(\hat\lambda^\star) \approx \frac{\mathrm{se}\big(\hat V_n(\lambda^\star)\big)}{\big|V_0'(\lambda^\star)\big|}. \tag{25}$$

With $n = 20{,}000$ paths, the standard error of the dirty price is 0.05–0.06 points and that of the implied hazard is 0.67bp for all three bonds (Appendix B). Because the three bonds are priced on the same paths, their Monte Carlo errors are strongly positively correlated, and the cross-bond dispersion of the implied hazards is much more precise than each hazard: across independent seeds and path counts from 5,000 to 40,000, the dispersion stays within 2.3–2.9bp.

4.4 Tier 2 and senior bonds

A Tier 2 or senior non-preferred note with coupons c_i to a call date T treated as certain, recovery R at resolution and resolution hazard $\lambda_{\text{res}} s_t^+/s_0$ pays its coupons and principal if $\tau_{\text{res}} > T$, and RF at τ_{res} if $\tau_{\text{res}} \le T$.

Proposition 4.4 (Tier 2 price)**.** *With recovery paid at the resolution time,*

$$B_0 = \mathbb{E}\left[\sum_{t_i \le T} c_i F\, e^{-rt_i - \lambda_{\text{res}} I_{t_i}} + F\, e^{-rT - \lambda_{\text{res}} I_T} + RF \int_0^T e^{-ru}\, \lambda_{\text{res}} \frac{s_u^+}{s_0}\, e^{-\lambda_{\text{res}} I_u}\, \mathrm{d}u\right]. \tag{26}$$

On a grid $0 = u_0 < u_1 < \cdots < u_N = T$ with recovery paid at the end of the step in which resolution occurs, the recovery leg is exactly

$$RF \sum_{n=0}^{N-1} \left(e^{-\lambda_{\text{res}} I_{u_n}} - e^{-\lambda_{\text{res}} I_{u_{n+1}}}\right) e^{-ru_{n+1}}, \tag{27}$$

and the difference from (26) *is bounded by $RF\big(1 - e^{-r\Delta}\big)$.*

Proof. The survival terms follow as in Theorem 4.1 with (19). For the recovery leg, conditionally on $\mathcal{G}_\infty$ the law of τ_{res} has density $\lambda_{\text{res}}(s_u^+/s_0)e^{-\lambda_{\text{res}} I_u}$, so $\mathbb{E}[RFe^{-r\tau_{\text{res}}}\mathbf{1}_{\{\tau_{\text{res}} \le T\}} \mid \mathcal{G}_\infty]$ is the integral in (26). On the grid, the conditional probability of resolution in $(u_n, u_{n+1}]$ is $e^{-\lambda_{\text{res}} I_{u_n}} - e^{-\lambda_{\text{res}} I_{u_{n+1}}}$, which gives (27). Paying at u_{n+1} instead of at the resolution time within the step changes the discount factor by at most a factor $e^{-r\Delta}$, and the total probability of resolution is at most one. □

With a monthly grid and $r \approx 5\%$ the bound is 0.4% of the recovered amount, that is about 0.01 points for a recovery of 20% and a resolution probability of about 13% to the call. The price (27) is again smooth and strictly decreasing in λ_{res} for $R < 1$, and the implied λ_{res} is unique. Crucially, the Tier 2 is priced on the *same* paths and the same driver I_t as the AT1s, so that λ_{res} and λ are directly comparable.

4.5 A term structure for the resolution hazard

A single level λ_{res} is calibrated to one Tier 2 bond. With two bonds of different maturities we can bootstrap a piecewise-constant level, $\lambda_{\text{res}}(t) = \lambda_1 \mathbf{1}\{t \le T^*\} + \lambda_2 \mathbf{1}\{t > T^*\}$, where T^* is the call of the shorter bond. The integrated hazard becomes $\int_0^t \lambda_{\text{res}}(u) \frac{s_u^+}{s_0}\, \mathrm{d}u = \lambda_1 I_{t \wedge T^*} + \lambda_2 \big(I_t - I_{T^*}\big)^+$.

Lemma 4.5. *Because I is non-decreasing, $I_{t \wedge T^*} = \min(I_t, I_{T^*})$ and $(I_t - I_{T^*})^+ = \max(I_t - I_{T^*}, 0)$. Hence every price under the piecewise hazard is computed from the stored values I_{t_i} together with the single path-wise value I_{T^*}.*

The bootstrap is sequential: λ_1 is implied from the shorter bond, whose cash flows end at T^*; λ_2 is then implied from the longer bond with λ_1 fixed; finally the AT1-specific level is implied from each AT1 with the piecewise resolution hazard in place. Each step is a one-dimensional root with the monotonicity of Proposition 4.3.

4.6 Discretisation

The state is simulated on a monthly grid ($\Delta = 1/12$) over $H = 40$ years with 20,000 paths. The spread uses its exact Gaussian transition,

$$S_{n+1} = \bar{s} + (S_n - \bar{s})e^{-a\Delta} + \eta \sqrt{\frac{1 - e^{-2a\Delta}}{2a}} \left(\rho Z_n^C + \sqrt{1 - \rho^2}\, Z_n^\perp\right), \tag{28}$$

where Z_n^C is the normal of the CET1 step and $Z_n^\perp$ an independent normal. The CET1 ratio uses an Euler step for the diffusion with a Bernoulli jump of probability $1 - e^{-\lambda_J \Delta}$ and lognormal size. The integrated driver uses the left-point rule, $I_{n+1} = I_n + \Delta\, s_n^+ / s_0$.

Three discretisation effects deserve comment. First, the Euler scheme for a linear SDE has weak order one; with $\kappa\Delta \approx 0.03$ the bias in the transition mean is of order $(\kappa\Delta)^2/2 \approx 5 \times 10^{-4}$ in relative terms. Second, the trigger is monitored only at grid dates, which underestimates the probability of crossing between dates. Broadie et al. (1997) show that, for a Brownian motion with volatility σ, discrete monitoring is approximately equivalent to shifting the barrier away by $0.5826\,\sigma\sqrt{\Delta}$; here the shift is 0.08pp of CET1 against a distance of 7.8pp to the trigger, and the conversion component of the spread is only 4bp, so the effect is negligible. Third, events on the grid are read at grid dates, while coupons are discounted at their exact dates. Halving the step to fortnightly changes the implied hazards by at most 1.6bp (Appendix B).

4.7 Common random numbers

All primitive random numbers are drawn once and reused across bonds, scenarios, bumps and feature switches. A difference between two prices computed on the same paths (a Greek, a stress P&L, a component of the spread decomposition, the dispersion of the implied hazards across bonds) therefore carries no Monte Carlo noise of its own beyond the second-order effect of the change on the path-wise payoffs. This is what makes the attribution of Section 5 exact on a given set of paths.

5 Attributing the spread

The market quotes AT1 bonds by their yield to the first call over the matching Treasury. We want to say how much of that spread pays for each risk. This section defines the attribution and discusses its dependence on the order in which risks are switched on.

5.1 Features, prices and yields

Index the risk features of the bond by the set

$$\mathcal{N} = \{\mathrm{R}, \mathrm{S}, \mathrm{M}, \mathrm{C}, \mathrm{E}\},$$

standing for resolution hazard (λ_{res}), AT1-specific hazard (λ_{spec}), MDA coupon restriction, conversion at the trigger, and extension (the economic call rule (13) instead of a certain call at T_1). For any subset $K \subseteq \mathcal{N}$ let $V(K)$ be the dirty price with exactly the features in K switched on, computed on the fixed set of paths:

- hazard $(\lambda_{\mathrm{res}} \mathbf{1}_{\mathrm{R}\in K} + \lambda_{\mathrm{spec}} \mathbf{1}_{\mathrm{S}\in K})\, s_t^+ / s_0$;
- coupons scaled by ϕ if $\mathrm{M} \in K$, paid in full otherwise;
- conversion at τ_m if $\mathrm{C} \in K$, no trigger otherwise;
- call by (13) if $\mathrm{E} \in K$, certain call at T_1 otherwise.

$V(\emptyset)$ is the price of a risk-free bullet to T_1, and $V(\mathcal{N})$ is the model price, equal to the market price by calibration.

Let $y(V)$ be the semi-annual yield to the first call implied by a dirty price V, treating the bond as a bullet to T_1, and define the spread of a subset as $v(K) = y(V(K)) - y(V(\emptyset))$. Because $V(\emptyset)$ is discounted at the risk-free rate, $y(V(\emptyset))$ is the Treasury yield and $v(\mathcal{N})$ is the market yield-to-call spread. The map $V \mapsto y(V)$ is strictly decreasing, so v is increasing in risk.

5.2 Sequential attribution and its order dependence

Given an order $\pi = (\pi_1, \ldots, \pi_5)$ of the features, the sequential attribution is

$$\varphi^{\pi}_{\pi_k} = v(\{\pi_1, \ldots, \pi_k\}) - v(\{\pi_1, \ldots, \pi_{k-1}\}), \tag{29}$$

which telescopes to $\sum_k \varphi^{\pi}_{\pi_k} = v(\mathcal{N})$: the components add up exactly to the market spread. The attribution depends on π whenever the features interact, that is whenever v is not additive. Interactions are pervasive here. The MDA cuts coupons that would otherwise be paid only if the bond survives, so its cost depends on the hazard; the value of the call option depends on the discount rate of the extended cash flows, so its cost depends on the hazard and on the MDA; and the yield transformation is non-linear.

The base attribution of the short note uses the order (R, S, M, C, E): the hazard first, so that every contract feature is valued with risky discounting, and extension last, so that the call is valued with every other feature in place.

5.3 Shapley values

The Shapley value (Shapley, 1953) averages the sequential attribution over all $|\mathcal{N}|! = 120$ orders:

$$\varphi^{\mathrm{Sh}}_j = \sum_{K \subseteq \mathcal{N} \setminus \{j\}} \frac{|K|!\,(|\mathcal{N}| - |K| - 1)!}{|\mathcal{N}|!} \big(v(K \cup \{j\}) - v(K)\big). \tag{30}$$

It is the unique attribution satisfying efficiency ($\sum_j \varphi_j = v(\mathcal{N})$), symmetry (interchangeable features receive the same value), the null-player property and additivity. It needs the $2^5 = 32$ values $v(K)$, which on a fixed set of paths require only eight cash-flow profiles per bond (one for each combination of M, C and E): the two hazard features enter through the discount exponent and are evaluated analytically.

The full Shapley value treats all orders as equally meaningful. For AT1 bonds they are not. Orders in which extension is switched on before any hazard value the issuer's option on a risk-free bond. When a risk-free bond pays its reset margin m over Treasuries, not calling it is a gift to the investor, so these orders assign extension a large negative value; this is precisely the inconsistency that the state-dependent hazard of Section 3.6 is designed to avoid. We therefore also compute a Shapley value within blocks.

Definition 5.1 (Nested Shapley value)**.** Partition the features into the hazard block $\mathcal{H} = \{\mathrm{R}, \mathrm{S}\}$ and the contract block $\mathcal{K} = \{\mathrm{M}, \mathrm{C}, \mathrm{E}\}$. The nested value is the Shapley value of the game v restricted to $\mathcal{H}$ for $j \in \mathcal{H}$, and the Shapley value of the game $K \mapsto v(\mathcal{H} \cup K) - v(\mathcal{H})$ on $\mathcal{K}$ for $j \in \mathcal{K}$.

The nested value is symmetric within each block and enters the hazard block first. It is related to the value for games with a priori unions of Owen (1977), which would also average over the order of the two blocks; we fix the order of the blocks for the economic reason just given. By construction it is efficient: the hazard block receives $v(\mathcal{H})$ and the contract block $v(\mathcal{N}) - v(\mathcal{H})$.

Proposition 5.2 (Split of the hazard block)**.** *Within the hazard block, the nested value is*

$$\varphi_{\mathrm{R}} = \tfrac{1}{2}\big[v(\{\mathrm{R}\}) + v(\mathcal{H}) - v(\{\mathrm{S}\})\big], \qquad \varphi_{\mathrm{S}} = \tfrac{1}{2}\big[v(\{\mathrm{S}\}) + v(\mathcal{H}) - v(\{\mathrm{R}\})\big].$$

If the spread were linear in the hazard level on the relevant range, $v(\{\mathrm{R}\}) + v(\{\mathrm{S}\}) = v(\mathcal{H})$ and the nested value would coincide with the sequential attribution in either order, giving $\varphi_{\mathrm{R}}/(\varphi_{\mathrm{R}} + \varphi_{\mathrm{S}}) = \lambda_{\mathrm{res}}/\lambda$.

Proof. The first statement is (30) for two players. Under linearity $v(\{\mathrm{R}\}) = \lambda_{\mathrm{res}}\, g$ and $v(\{\mathrm{S}\}) = \lambda_{\mathrm{spec}}\, g$ for some slope g, and both orders give the same increments. □

For BNP the yield-to-call spread is very close to linear in the hazard on the relevant range, so the split of the hazard block is effectively order-free: the nested value and the base order agree to within 0.2bp (Section 8). The order matters for the contract block, and there the attribution should be reported as a range.

6 Data and calibration

6.1 Data

Bond prices. Daily Capital IQ evaluated bid, mid and ask prices for the three AT1s, the two Tier 2 notes and the SNP note, from each bond's issue date (the earliest being 27 June 2025) to 25 September 2026. Grey-market quotes before the issue date are discarded, as are weekend repetitions of Friday's evaluation. The analysis uses weekly (Friday) observations, 66 dates. Accrued interest is computed on a 30/360 basis on the trade date and matches Capital IQ's "T0 accrued interest" to three decimals. Evaluated prices are composites of dealer quotes rather than transactions; their bid–offer on the valuation date is 0.15–0.29 points for the AT1s and 0.07–0.17 for the Tier 2 and SNP notes, all of which carry Capital IQ's highest liquidity score.

Treasury curve. FRED constant-maturity Treasury yields at five, seven and ten years (DGS5, DGS7, DGS10). The five-year CMT is the reset reference of the AT1s; each bond is discounted at the yield interpolated linearly at its time to first call, flat outside the five-to-ten-year range.

Capital. Quarterly CET1 ratios from Capital IQ, 2014Q1 to 2026Q2. Earlier data follow Basel II/2.5 definitions and are not comparable. A print is used only from 35 days after the quarter end, approximating BNP's publication lag. The CET1 requirement is the year-end stack from Capital IQ up to 2024 and the quarterly figures published by BNP thereafter (10.51% at September 2025, 10.43% at June 2026).

Credit default swaps. BNP Paribas senior CDS, EUR, MM14 documentation: the five-year conventional spread daily since October 2023 and full curves on six snapshot dates. The CDS is used only as a reference point, because it is quoted in a different currency and is referenced to senior rather than subordinated debt.

Other. BNP's share price (EUR) and the EBA 2025 stress test results, in which BNP's fully loaded CET1 ratio falls by 4.21pp in the adverse scenario.

6.2 Capital dynamics

Likelihood. Ignoring jumps, the CET1 process is an Ornstein–Uhlenbeck process whose transition over a step Δ_k is Gaussian:

$$C_{k+1} \mid C_k \sim N\big(m_k,\ \sigma^2 w_k\big), \qquad m_k = \theta + (C_k - \theta)e^{-\kappa\Delta_k}, \qquad w_k = \frac{1 - e^{-2\kappa\Delta_k}}{2\kappa}. \tag{31}$$

The exact log-likelihood of n transitions is

$$\ell(\kappa, \sigma) = -\frac{1}{2}\sum_k \Big[\log\big(2\pi\sigma^2 w_k\big) + \frac{(C_{k+1} - m_k)^2}{\sigma^2 w_k}\Big]. \tag{32}$$

For fixed κ the maximiser in σ is available in closed form,

$$\hat\sigma^2(\kappa) = \frac{1}{n}\sum_k \frac{(C_{k+1} - m_k)^2}{w_k}, \tag{33}$$

which gives the profile likelihood $\ell_p(\kappa) = -\frac{n}{2}\big[1 + \log(2\pi\hat\sigma^2(\kappa))\big] - \frac{1}{2}\sum_k \log w_k$, maximised numerically in κ. The long-run level is fixed at $\theta = 13\%$, BNP's communicated target; with fifty observations of a slowly mean-reverting series, a free θ would be poorly determined and would interact with κ. The 2023Q1 transition (+1.2pp, the sale of Bank of the West) is a disposal, not a capital shock, and is excluded from the sum.

Estimates. The maximum likelihood estimates are $\hat\kappa = 0.165$ and $\hat\sigma = 0.44\text{pp}/\sqrt{\text{yr}}$, a half-life of a capital shock of $\ln 2/\hat\kappa = 4.2$ years. Figure 2 shows the history, the requirement and the base-case outlook.

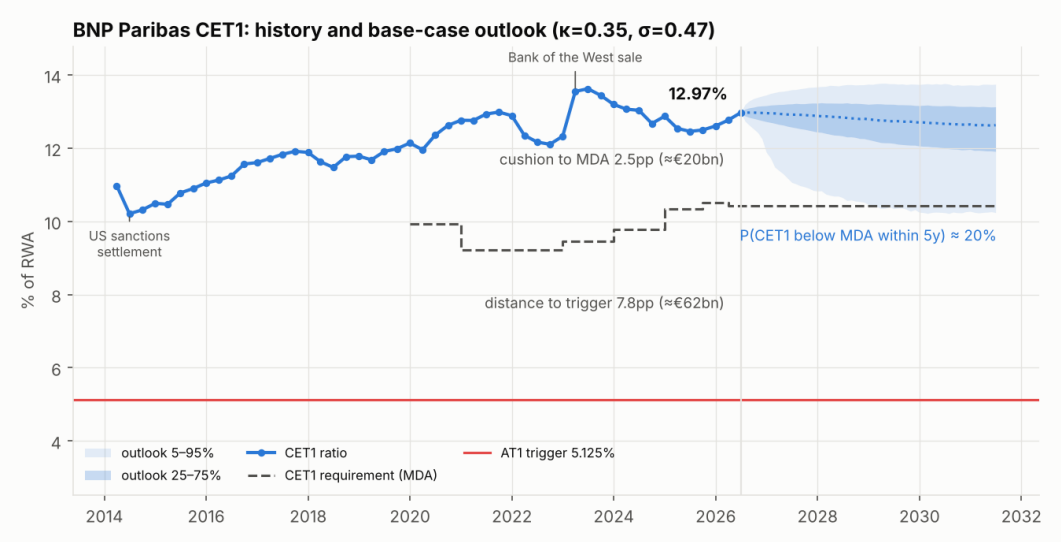


Figure 2: BNP Paribas CET1 ratio since 2014, CET1 requirement (MDA threshold), AT1 trigger and the base-case five-year outlook (median, 50% and 90% bands of simulated CET1).

Bootstrap. To measure the precision of $\hat\kappa$ we simulate 1,000 histories of the fitted process on the observed quarterly dates and re-estimate (κ, σ) on each. The 90% intervals are $\kappa \in [0.063, 0.476]$ and $\sigma \in [0.37, 0.52]$: the volatility is well identified, the speed of capital rebuilding is not. The

half-life of a capital shock lies between 1.5 and 11 years. This is the familiar difficulty of estimating mean reversion from a short sample of a persistent series, and it matters: Section 7.7 shows that the capital stress results scale by a factor of five across the interval.

Base case. The base case uses $\kappa = 0.35$ (half-life two years), inside the bootstrap interval, with $\sigma = 0.47$ re-estimated by (33). The likelihood ratio statistic against the maximum likelihood estimate is $2 \times 1.5 = 3.0$, below the 5% critical value of 3.84 for one restriction. Section 7.7 explains the choice: bond prices lean towards faster mean reversion, and BNP's explicit capital management (a 13% target with managed risk-weighted assets and distributions) is consistent with it.

Jumps. The jump parameters cannot be identified from fifty quarters with few large shocks. We use $\lambda_J = 0.12$ a year, $m_J = 2.2$pp and $\sigma_J = 0.5$. The 95th percentile of the jump size, $m_J e^{1.645\sigma_J - \sigma_J^2/2} = 4.4$pp, matches the 4.2pp CET1 depletion of BNP in the EBA 2025 adverse scenario: the calibration says that a shock of the size of the adverse scenario occurs as a single jump with probability $0.12 \times 5\% = 0.6\%$ a year, about once every 170 years; smaller jumps are more frequent and can accumulate.

6.3 Spread process and regulatory geometry

The market component of the refinancing spread starts on each date at the average, across the AT1s quoted that day, of the yield to first call minus the call-matched Treasury. It mean-reverts at $a = 0.5$ (half-life 1.4 years) to the sample mean $\bar{s} = 275$bp, with volatility $\eta = 100$bp and correlation $\rho = -0.3$ with capital shocks. The capital premium is $\beta = 1\%$ of spread per point of CET1 inside the buffer. Neither η nor β is identified by the data; Section 7 reports the sensitivity of the cross-section to η, and β is discussed in Section 9. The MDA geometry is as in (11), the trigger is 5.125%, and the stressed share price at conversion is $\xi = 20\%$ of today's.

6.4 Implied hazards and the cross-sectional test

On each date and for each AT1 j we solve $V_0^{(j)}(\lambda_j) = P_j^{\text{clean}} + \text{AI}_j$. The three bonds share issuer, capital and trigger, so a correctly specified model should give $\lambda_1 = \lambda_2 = \lambda_3$, and the dispersion

$$D = \max_j \lambda_j - \min_j \lambda_j$$

is a specification test. It can also be used to ask which values of the unidentified parameters (η, κ) the three prices are consistent with.

Power of the test. The test has limited power with evaluated prices. By Proposition 4.3, a difference $\Delta\lambda$ in hazard corresponds to a price difference of $|V_0'|\Delta\lambda$, with $|V_0'| = 0.077$–0.084 points per basis point. The bid–offer of 0.15–0.29 points therefore corresponds to 1.9–3.5bp of hazard. A dispersion of a few basis points is within the precision of the data, and the test can reject specifications that produce large dispersions but cannot rank specifications whose dispersions differ by one or two basis points. Monte Carlo error is not the binding constraint: the standard error of each implied hazard is 0.67bp, and the dispersion, computed on common random numbers, is more precise still.

6.5 Tier 2 calibration and the identification of recovery

On each date the Tier 2 and SNP notes are priced on the same simulated state as the AT1s, with the spread process started at the same AT1 spread and the same CET1 print, and with recoveries $R_{\mathrm{T2}} = 20\%$ and $R_{\mathrm{SNP}} = 40\%$, the Markit conventions for subordinated and senior debt. Each bond gives an implied resolution hazard by Proposition 4.4; the Tier 2 2035NC34 is the anchor for λ_{res}.

Recovery and hazard are not separately identified from a single bond: a higher recovery requires a higher hazard to match the same price. With two bonds of the same tenor and different seniority, Tier 2 and SNP, the restriction that both face the same resolution hazard identifies one recovery given the other. We report the Tier 2 recovery $R^{\star}$ that equalises the two implied hazards given $R_{\mathrm{SNP}} = 40\%$, and the sensitivity of the AT1 split to R_{T2}.

7 Results: BNP Paribas, 25 September 2026

All results are as of 25 September 2026. In the following month BNP's spreads widened with French sovereign stress, the ten-year OAT yield rising above 5%: the senior CDS five-year spread rose from 57 to 74bp and, in Capital IQ spreads to benchmark, the SNP from 98 to 122bp and the Tier 2 2035NC34 from 141 to 153bp by 27 October. The analysis does not include that move.

7.1 Capital

CET1 was 12.97% at June 2026 against a 10.43% requirement: a cushion of 2.54pp, about EUR 20bn of losses at June 2026 risk-weighted assets of EUR 795bn. The trigger is 7.8pp below the ratio, about EUR 62bn of losses. Under the base-case dynamics, 20% of the simulated paths fall below the current MDA threshold at some point in the next five years, while the trigger is almost never reached by diffusion alone. For these bonds capital risk is first a coupon and call risk, and only then a conversion risk.

7.2 The three bonds and the implied hazard

Table 2 reports the implied hazards on the valuation date. Without write-down risk the model would value the bonds at 114–118, against market prices of 94–98: the mechanical features alone explain only a small part of the discount. The implied hazards are 2.21%, 2.18% and 2.19% a year.

	6.875% NC33	7.20% NC36	7.45% NC35
Reset margin (bp)	285	294	313
Clean price	94.39	95.25	97.57
Bid–offer (points)	0.26	0.29	0.15
Yield-to-call spread (bp)	279	274	269
Model price with $\lambda = 0$	114.2	116.9	117.6
Implied hazard λ (%/yr)	2.21	2.18	2.19
Monte Carlo s.e. of λ (bp)	0.67	0.66	0.67
Pr(called at T_1)	49%	52%	58%
Pr(called at some reset)	97%	98%	98%
Lifetime Pr(conversion)	1.4%	1.6%	1.4%

Table 2: Implied hazard and model outputs on 25 September 2026, base case ($\kappa = 0.35$, $\eta = 100$bp, economic call rule).

Cross-section. Table 3 shows the dispersion of the three implied hazards as a function of the spread volatility η. A deterministic refinancing spread ($\eta = 0$) gives a dispersion of 8.2bp; a stochastic one reduces it steadily, without an interior minimum, so the cross-section does not identify η and 100bp is kept as an assumption. In price terms the base-case dispersion of 2.6bp is 0.20–0.22 points, of the size of the bid–offer: the bonds are consistent with one hazard within the precision of the data.

η (bp)	0	25	50	75	100	150	200
Dispersion $D(\eta)$ (bp of hazard)	8.2	5.8	4.1	3.0	**2.6**	2.0	1.4
In price points (6.875%)	0.63	0.45	0.32	0.23	**0.20**	0.15	0.11

Table 3: Cross-bond dispersion of the implied hazards on the valuation date, by refinancing-spread volatility.

Over the 66 weekly dates the mean dispersion is 5.9bp with the stochastic spread and 9.3bp with the deterministic one. Priced at the day's common hazard, each bond's average pricing error is within ±0.25 points (range −1.1 to +1.1). Using Capital IQ benchmark yields instead of the FRED curve, the last-date dispersion is 2.3–7.8bp depending on the benchmark assignment, so the choice of risk-free curve matters as much as the choice of η. Figure 3 shows the implied hazards over time. The average hazard ranged from 1.82% (August 2026) to 2.69% (late March 2026, when the bonds then outstanding fell four to five points) and rose again into the September sell-off.

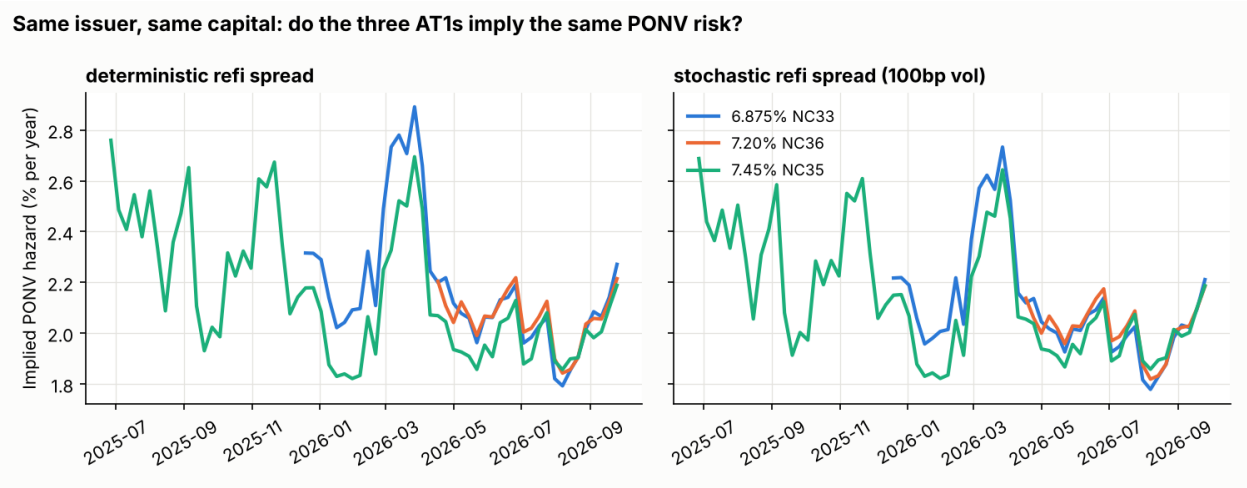


Figure 3: Implied hazard per bond, weekly, with a deterministic (left) and a stochastic (right) refinancing spread.

7.3 What BNP's Tier 2 prices

Table 4 sets the AT1s against the rest of BNP's capital structure. Bond hazards are implied on the same simulated state as the AT1s; CDS hazards use the credit triangle $\lambda \approx s/(1-R)$.

	Spread (bp)	Recovery	Hazard (%/yr)	Share of AT1 hazard
Senior CDS 5Y (EUR)	57	40%	0.95	43%
Senior CDS 7Y (EUR, 27 Sep)	68	40%	1.13	52%
Senior CDS 10Y (EUR, 27 Sep)	81	40%	1.35	62%
SNP 5.738% 2035NC34	103	40%	1.63	74%
T2 2.588% 2035NC30	116	20%	1.40	64%
T2 5.906% 2035NC34 (anchor)	**144**	**20%**	**1.72**	**78%**
AT1 (average of three)	274	0%	2.19	100%

Table 4: BNP Paribas's capital structure on 25 September 2026. Bond spreads are yield to call over the call-matched FRED Treasury.

The ladder is orderly: each step down the capital structure pays more. The two instruments that share the resolution event, the Tier 2 and the SNP of the same tenor, imply nearly the same hazard (1.72% and 1.63%) under the Markit recoveries. With the SNP at 40%, the Tier 2 recovery that equalises the two is $R^\star = 15.6\%$, close to the conventional 20%: the prices of BNP's own bonds are consistent with the recovery conventions. The CDS sits below the bonds, as one would expect of a EUR contract on senior debt compared with USD cash bonds that carry funding and liquidity premia. The shorter Tier 2 implies a lower hazard than the longer one, an upward-sloping term structure that Section 8 exploits.

With the Tier 2 2035NC34 as anchor,

$$\lambda = 2.19\% = \underbrace{1.72\%}_{\lambda_{\text{res}}\text{: priced in Tier 2}} + \underbrace{0.47\%}_{\lambda_{\text{spec}}\text{: AT1-specific}} . \tag{34}$$

By Proposition 3.5, 78% of the AT1's write-down risk, in risk-neutral terms, is compensation that BNP's Tier 2 also earns; the remaining 22% is what the market charges for being an AT1 rather than a Tier 2. Over the 66 weekly dates the Tier 2 share ranges from 66% to 80%, with a mean of 72% (Figure 4).

Tier 2 recovery	0%	10%	20%	30%	40%
λ_{res} (%/yr)	1.37	1.53	1.72	1.97	2.30
λ_{spec} (%/yr)	0.82	0.67	0.47	0.23	−0.11
Tier 2 share of the AT1 hazard	63%	70%	78%	90%	105%

Table 5: Sensitivity of the split to the Tier 2 recovery.

The split depends on the Tier 2 recovery (Table 5). A recovery as high as the senior one would leave a negative AT1-specific hazard, which is implausible; any recovery up to 30% attributes at least three-fifths of the AT1 hazard to what Tier 2 also prices.

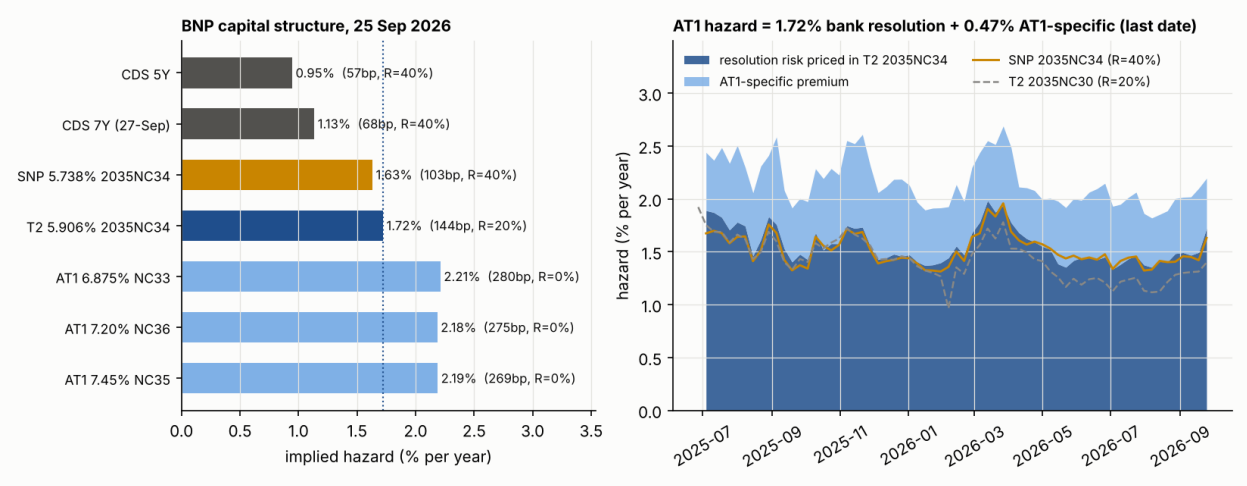


Figure 4: Left: implied hazards down BNP's capital structure. Right: the AT1 hazard (average of the three bonds), weekly, split into the part priced in the Tier 2 2035NC34 and the AT1-specific premium; the SNP and the shorter Tier 2 are shown for comparison.

7.4 What the spread pays for

Table 6 and Figure 5 give the base attribution of the yield-to-call spread (order R, S, M, C, E).

bp of yield-to-call spread	6.875% NC33	7.20% NC36	7.45% NC35
Priced in Tier 2 (resolution)	181	181	180
AT1-specific premium	52	49	49
MDA coupon cuts	27	31	30
Conversion at the trigger	4	4	4
Extension	15	9	6
Total = market	279	274	269

Table 6: Attribution of the yield-to-call spread, base order. Components are rounded to sum to the total.

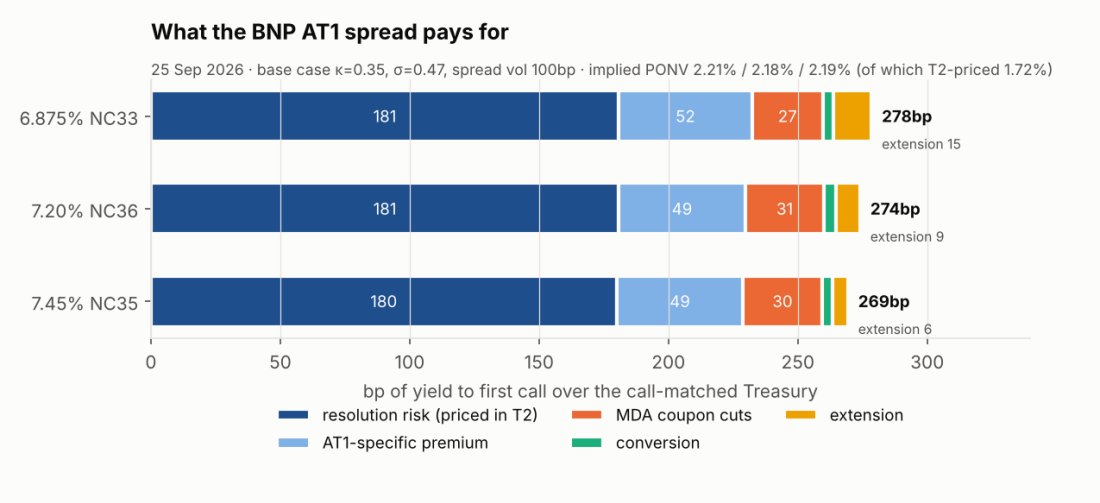


Figure 5: What the BNP AT1 spread pays for.

Two-thirds of the spread, about 180bp, is priced in Tier 2 as well. Another 50bp is the premium for being an AT1 rather than a Tier 2: the risk of a write-down without resolution, the equity-like, perpetual and discretionary profile, and the instrument's liquidity. The contract features explain the rest. Expected MDA coupon cuts cost about 30bp. Conversion at the trigger, a 1.4–1.6% lifetime event at a recovery of about a third, costs 4bp. Extension costs 6–15bp and decreases with the reset margin, as it should: the 6.875% has the lowest margin over today's AT1 spread and pays most for the issuer's option.

The MDA component is an upper bound, for the reasons given in Section 2: the model applies the quartile factor to the coupon, while a bank inside its buffer would cut dividends, buy-backs and variable pay first. Because the total spread is fixed by the market price, an overstated MDA component is taken from the AT1-specific premium, so the 50bp is a lower bound.

7.5 Call and extension

	6.875% NC33	7.20% NC36	7.45% NC35
Reset margin − today's AT1 spread (bp)	+6	+15	+34
Pr(called at T_1)	49%	52%	58%
Pr(called at some reset)	97%	98%	98%
Cost of the issuer's option (points)	0.78	0.59	0.37
P&L if AT1 spreads settle +50bp (points)	−3.3	−3.5	−3.2
of which from extension (points)	−1.2	−0.9	−0.7

Table 7: Call and extension. The cost of the option is the price if the call at T_1 were certain minus the model price, at the same hazard. The extension P&L is the extra loss relative to a bond whose call at T_1 is certain.

With a purely economic rule the first call is close to a coin toss, and almost certain at some later reset. Figure 6 shows the probability of a call at the first reset as a function of the long-run AT1 spread: at 200bp all three bonds are called about three times out of four, at 350bp only a quarter

to a third of the time. The low-margin 6.875% is always the least likely to be called and is the most negatively convex: if AT1 spreads settle 50bp wider, it loses 1.2 points more than a bond with a certain call. The market priced exactly this in September 2026: with the ten-year Treasury up 45bp in four weeks, the 6.875%'s spread widened by a further 34bp, against 10–12bp for the other two.

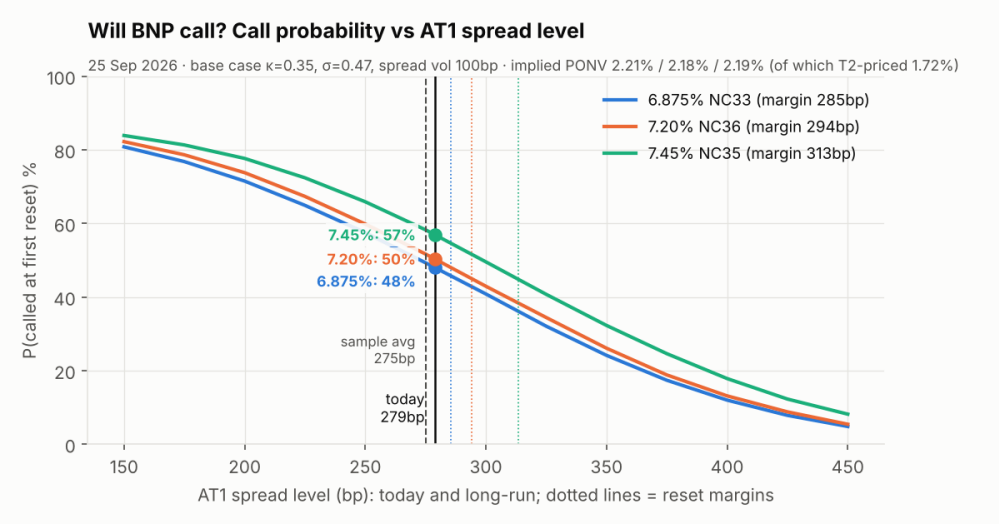


Figure 6: Probability of a call at the first reset against the long-run AT1 spread level.

7.6 Stress and the capital cliff

Each scenario of Table 8 reprices the bonds on the same random numbers. Rate shocks move the discount rate and the reset reference; spread shocks move the current and long-run AT1 spread and scale the hazard with it; capital shocks move today's CET1 ratio. The two hazard shocks have market counterparts: doubling the Tier 2-priced hazard is roughly what a doubling of BNP's Tier 2 spread would imply, and it costs three times as much as doubling the AT1-specific premium.

Clean price change (points)	6.875% NC33	7.20% NC36	7.45% NC35
Rates +100bp	−5.3	−6.4	−6.1
Rates −100bp	+5.7	+7.0	+6.7
CET1 −2pp	−1.5	−1.6	−1.6
CET1 −4.2pp (EBA adverse)	−9.8	−11.1	−10.4
AT1 spreads +200bp	−16.4	−16.2	−15.4
Spreads +200bp, rates −50bp (March 2023 style)	−14.5	−13.8	−13.0
EBA adverse + spreads +200bp	−25.4	−26.6	−25.1
Tier 2-priced hazard ×2	−11.9	−13.0	−12.3
AT1-specific premium ×2	−3.6	−3.8	−3.6

Table 8: Stress scenarios, base case.

CET1 level	13% (today)	12%	11%	10.5%	9%	7%
6.875% NC33	0.6	0.8	1.7	4.4	4.6	4.6
7.45% NC35	0.6	0.8	1.9	4.8	4.8	4.8

Table 9: Capital delta: price points per +1pp of CET1, by CET1 level (finite differences on a 0.5pp grid).

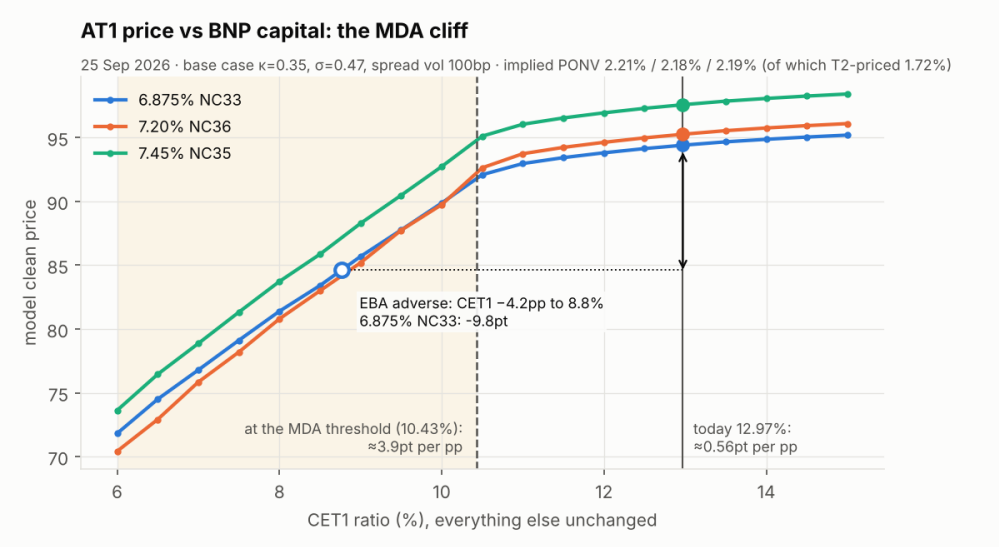


Figure 7: Model price against today's CET1 ratio, everything else unchanged.

Table 9 and Figure 7 show the capital cliff. At today's ratio the price moves by about 0.6 points per point of CET1; within a point of the MDA threshold the sensitivity rises to 4–5 points. Crossing the threshold switches on three mechanisms at once: coupons are cut by (12), the call is blocked by (13), and the hazard rises through (8)–(15). Because the O-SII step-ups raise the threshold by 50bp by 2028, a capital loss of about half the EBA adverse depletion moves the bonds from the flat part of the curve onto the cliff. The relevant risk measure for these bonds is the distance to the MDA threshold, not the distance to the trigger.

7.7 How much can we trust the capital parameters?

Table 10 reports the headline results, averaged across the three bonds, at the two ends of the bootstrap interval for κ, at the maximum likelihood estimate and in the base case. Each column recalibrates the implied hazards.

	$\kappa = 0.06$ (5%)	$\kappa = 0.165$ (MLE)	$\kappa = 0.35$ (base)	$\kappa = 0.48$ (95%)
Half-life of a capital shock (years)	11.0	4.2	2.0	1.5
Implied hazard (%/yr)	0.7	1.6	2.2	2.3
MDA coupon cost (points)	14.5	6.3	2.5	1.7
Capital delta at the MDA threshold (points/pp)	13.7	8.8	4.2	2.8
EBA adverse shock (points)	−37.7	−21.4	−10.5	−7.3
Cross-bond dispersion D (bp)	7.8	4.2	2.6	2.3
in price points	0.6	0.3	0.2	0.2
Log-likelihood relative to MLE	−0.5	0	−1.5	−4.1

Table 10: Sensitivity to the speed of capital mean reversion (average of the three bonds). The capital delta at the threshold is the central difference over ±0.25pp.

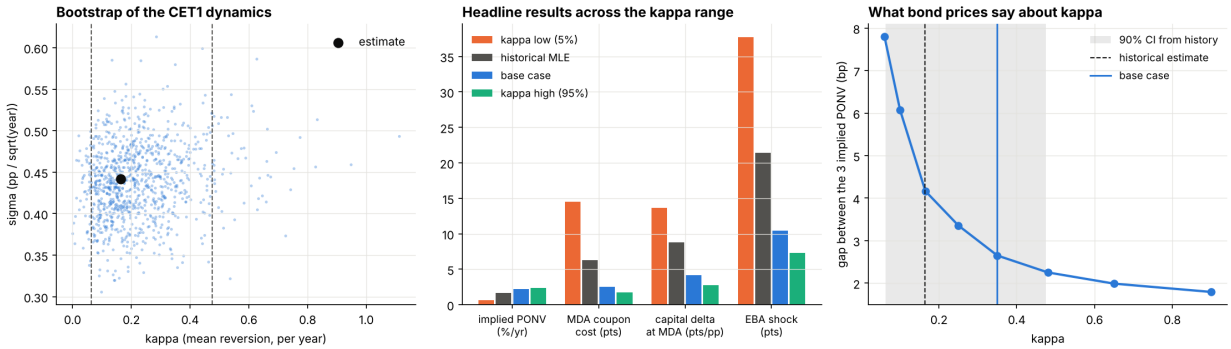


Figure 8: Left: bootstrap of (κ, σ). Centre: headline results across the κ range. Right: cross-bond dispersion of the implied hazard against κ.

What prices add. History and prices pull in opposite directions. The three bonds are more consistent with fast capital rebuilding: the dispersion falls from 7.8bp at $\kappa = 0.06$ to 2.3bp at 0.48 and then flattens. In price terms this is 0.6 against 0.2 points, around the bid–offer, so the evidence is a lean, not an identification. The history disfavours values much above 0.35: the likelihood ratio statistic is 3.0 at 0.35 and 8.2 at 0.48. The base case sits where the two sources of information overlap, and is consistent with BNP's active capital management.

Where the uncertainty sits. With a slower mean reversion a capital shock lasts longer, so more paths spend time inside the buffer: the MDA cost and the capital delta rise and, since the total is fixed by the price, the implied hazard falls. Figure 9 re-calibrates each curve to today's price of the 6.875%. All four agree at today's 13% CET1, and diverge only as capital falls: at the MDA threshold the price ranges from 76 ($\kappa = 0.06$) to 93 ($\kappa = 0.48$), and at 6% CET1 from 36 to 77. Bond prices identify the level of risk today, not the shape of the cliff. What is robust across the interval is the ordering of the three bonds, the extension ranking and the steepening of the capital delta near the threshold; the level of the capital stress and the split between the AT1-specific premium and the MDA component are not.

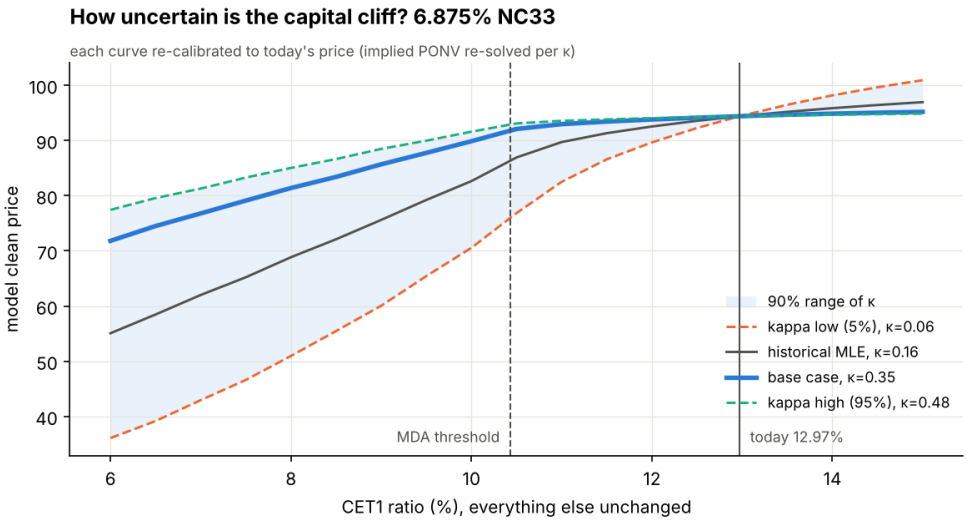


Figure 9: Price of the 6.875% NC33 against CET1 for four values of κ, each re-calibrated to today's price. Shaded: 90% bootstrap range of κ.

8 Robustness

This section asks whether the central result, the split of the AT1 spread into about 180bp priced in Tier 2 and 50bp of AT1-specific premium, survives three objections a practitioner would raise: that the call rule is unrealistic, that the attribution depends on the order of features, and that a single flat resolution hazard is too crude for a perpetual. All computations are on the valuation date, on the same random numbers as the base case.

8.1 Reputational calls

The base call rule (13) with $\delta = 0$ calls only when refinancing is cheaper. European national champions have almost always called their AT1s at the first opportunity, and market participants quote call probabilities well above those of Table 7. We introduce a reputational tolerance: the issuer also calls when its reset margin is up to δ *below* the new-issue spread. For each δ we recalibrate the hazard of each bond.

Tolerance δ (bp)	0	25	50	100
Pr(called at T_1): 6.875% NC33	49%	57%	65%	77%
7.20% NC36	52%	60%	67%	79%
7.45% NC35	58%	66%	72%	81%
Implied hazard, 6.875% (%/yr)	2.21	2.22	2.23	2.25
Implied hazard, 7.20% (%/yr)	2.18	2.19	2.20	2.21
Implied hazard, 7.45% (%/yr)	2.19	2.19	2.19	2.19
Extension component, 6.875% (bp)	15	14	13	11
Extension component, 7.20% (bp)	9	9	8	7
Extension component, 7.45% (bp)	6	6	6	6
Tier 2 share of the AT1 hazard	78%	78%	78%	78%

Table 11: Reputational call tolerance. The extension component is the yield-to-call spread of the model price minus that of the same bond with a certain call at T_1.

A tolerance of 100bp raises the probability of a call at the first reset to 77–81%, the range practitioners quote, and lowers the extension component of the low-margin bond from 15 to 11bp. The implied hazard moves by at most 4bp, and the Tier 2 share of the hazard does not move. The reason is that the market price pins down the total value of the bond: a more generous call rule raises the value of the call leg slightly, and the hazard adjusts to absorb it. The extension risk of the low-margin bond, and its negative convexity, are therefore partly a feature of the economic call rule, while the split between Tier 2-priced and AT1-specific risk is not.

8.2 Order of attribution

Table 12 reports four attributions of the spread for each bond: the base order (R, S, M, C, E), the nested Shapley value of Section 5, the full Shapley value over all 120 orders, and the reverse order (E, C, M, S, R).

Bond	Component	Base order	Nested Shapley	Full Shapley	Reverse order
6.875% NC33	Priced in Tier 2	180.7	180.8	221.1	259.1
	AT1-specific	51.6	51.6	62.8	77.0
	MDA	27.3	36.1	35.5	43.3
	Conversion	4.1	5.5	6.4	11.7
	Extension	14.7	4.5	−47.3	−112.6
7.20% NC36	Priced in Tier 2	180.8	180.9	206.9	231.4
	AT1-specific	48.8	48.8	55.7	64.4
	MDA	30.6	36.0	35.1	39.3
	Conversion	4.6	6.0	6.8	11.0
	Extension	9.1	2.4	−30.4	−72.0
7.45% NC35	Priced in Tier 2	180.0	180.0	203.6	225.9
	AT1-specific	48.9	48.8	55.1	62.9
	MDA	30.4	35.8	35.0	39.2
	Conversion	4.0	5.3	6.1	10.1
	Extension	6.0	−0.6	−30.4	−68.8

Table 12: Attribution of the yield-to-call spread (bp) under four methods. Totals are 278.5, 274.0 and 269.3bp in every column.

Three conclusions follow.

First, the split of the hazard block is order-free. In every attribution that prices the hazard before the contract features, the base order and the nested Shapley value, the Tier 2-priced and AT1-specific components agree to within 0.2bp. This is the near-linearity of Proposition 5.2: on the relevant range the yield-to-call spread is almost linear in the hazard level, so it does not matter in which order λ_{res} and λ_{spec} are switched on.

Second, the attribution among the contract features is order-dependent. Their total is the same in the base order and in the nested value (40–46bp depending on the bond), but its split is not: symmetric treatment raises the MDA component to about 36bp and reduces extension to a few basis points. The two features interact through the capital condition of the call rule. The states in which the call is blocked because CET1 is below the threshold are the states in which the MDA cuts coupons; once the MDA is priced, the cash flows of the bond in those states are already reduced, so the incremental cost of not being called is smaller, and conversely. The honest statement is that the contract features cost 40–46bp, of which MDA is 27–36bp, conversion 4–6bp and extension between −1 and 15bp.

Third, orders that switch extension on before the hazard produce economically meaningless results. Without any hazard, the bond is risk-free, and not calling a risk-free bond that pays its reset margin over Treasuries is a gift to the investor: extension takes a large negative value (−69 to −113bp in the reverse order) and the hazard components absorb the difference. This is the inconsistency the state-dependent hazard is designed to avoid (Section 3.6); it shows why the full Shapley value, which averages over such orders, is the wrong benchmark for AT1 attribution.

8.3 Term structure of the resolution hazard

A single flat λ_{res}, calibrated to a bond with a 2034 call, is applied to AT1 cash flows that run for decades if the bond is extended. With the two Tier 2 bonds we bootstrap a two-piece resolution hazard as in Section 4.5.

	6.875% NC33	7.20% NC36	7.45% NC35
λ_1 to August 2030 (%/yr)	1.40	1.40	1.40
λ_2 after August 2030 (%/yr)	2.10	2.10	2.10
Flat λ_{res} (%/yr)	1.72	1.72	1.72
λ_{spec}, flat λ_{res} (%/yr)	0.49	0.46	0.47
λ_{spec}, two-piece λ_{res} (%/yr)	0.42	0.36	0.39
Tier 2-priced component, flat (bp)	180.7	180.8	180.0
Tier 2-priced component, two-piece (bp)	176.4	184.6	181.2

Table 13: Two-piece resolution hazard bootstrapped from the Tier 2 2035NC30 (call August 2030) and the Tier 2 2035NC34 (call November 2034), recovery 20%.

The resolution hazard is upward sloping: 1.40% a year to 2030 and 2.10% after. With the steeper curve, more of each AT1's hazard is attributed to what Tier 2 prices, and the AT1-specific hazard falls from 0.47% to 0.39% on average. The Tier 2-priced component of the spread moves by less than 5bp, in different directions for different bonds depending on the timing of their cash flows. A term structure therefore strengthens, rather than weakens, the conclusion that most of the AT1

spread is priced in Tier 2. A caveat applies to the short Tier 2: it trades well below par with a reset in 2030, and its price may contain a convexity or tax effect that a bullet-to-call treatment ignores.

9 Discussion

9.1 What the AT1-specific premium is

The AT1-specific hazard, 0.4–0.5% a year, about 50bp of spread, is the part of the AT1's compensation that BNP's Tier 2 does not earn. Proposition 3.5 gives it a precise risk-neutral meaning: it is the intensity of write-downs that hit the AT1 without a resolution that also hits Tier 2, plus whatever premium investors attach to the instrument beyond its expected loss. Four components are plausible, and the model cannot separate them.

1. *Write-down without resolution.* The Credit Suisse case showed that AT1 holders can lose everything in a transaction that leaves more junior or more senior claims partially intact. The EU authorities' March 2023 statement makes this less likely in the EU, but it does not make it impossible, and investors price the tail.

2. *Coupon discretion beyond the MDA.* The issuer may cancel coupons for reasons other than the MDA, and supervisors can restrict distributions outside the MDA rule, as they did with dividends in 2020. The model's MDA rule does not capture this.

3. *Equity-like risk premium.* An AT1 bond is perpetual, its payoff in bad states is close to that of equity, and its investor base includes funds with mandates that force sales in stress. The premium for bearing this risk need not be the same as Tier 2's.

4. *Liquidity.* AT1 bonds trade with wider bid–offer spreads than Tier 2 and are excluded from some benchmark indices.

Because the MDA component is an upper bound, the 50bp is a lower bound for these four together.

9.2 Implications

For investors. Two-thirds of an AT1's spread is compensation that the same issuer's Tier 2 also pays, at a better recovery and without extension or coupon risk. The relevant relative-value question is therefore whether the AT1-specific 50bp, plus the 40–46bp of contract features, is adequate compensation for those features, not whether 275bp is a high spread in absolute terms. The model provides a consistent way to make that comparison across issuers: calibrate λ_{res} to each issuer's Tier 2 and compare the residuals.

For risk management. The price is flat in capital at today's level and steep within a point of the MDA threshold. A risk system that measures the distance to the trigger will understate the capital risk of these bonds; the relevant distance is to the threshold, which itself moves with scheduled buffer changes. The capital delta at the threshold is three to eight times the delta today, depending on the speed of capital mean reversion (Table 10).

For the extension decision. The low-margin bond is the most exposed to extension under an economic call rule, and the market behaved accordingly in September 2026. Whether the issuer

would in fact extend is a question about its reputational incentives; the robustness results show that this question affects the extension component of the spread, but not the hazard or its split.

9.3 Limitations

1. **Measure.** CET1 dynamics are estimated under the physical measure and assumed to hold under the pricing measure. All residual premia are in λ, which is a risk-neutral intensity. The Tier 2 anchor splits it, but λ_{res} is risk-neutral too and contains Tier 2's own premia.

2. **Recovery.** The split depends on the Tier 2 recovery (Table 5). The market prices are consistent with the conventional 20%, but recovery is not observed.

3. **Capital mean reversion.** κ is weakly identified, and the capital stress results scale by a factor of five across its bootstrap interval.

4. **MDA mechanics.** Applying the quartile factor to the coupon overstates the MDA cost. The leverage-ratio and MREL MDAs, the available distributable items and supervisory discretion over distributions are not modelled.

5. **Call behaviour.** The base rule is economic; a reputational tolerance changes the call probabilities but not the hazard. The regulatory approval test of CRR Art. 77–78 is approximated by the MDA threshold.

6. **No systemic or sovereign factor.** The spread process is issuer-specific and capital jumps do not move spreads. AT1 spreads widened by hundreds of basis points in March 2020 and March 2023 with no change in capital, and BNP's spreads widened with French sovereign stress after the valuation date. A common sector and sovereign factor with co-jumps in capital and spreads is the main extension.

7. **Rates.** The current curve is used for all future resets; a short-rate model would add rate-driven extension.

8. **Unidentified spread parameters.** The spread volatility η and the capital premium β are assumed. A cross-section of issuers, or a regression of AT1 spreads on the distance to the MDA threshold, would calibrate β.

9. **Data.** Prices are evaluated rather than traded; the sample covers fifteen months of relatively tight spreads; the bid–offer is of the same size as the cross-bond dispersion; the CDS is in EUR and the bonds in USD. TRACE transaction prices would allow a test on traded data.

10. **Conversion.** A single stressed share price replaces a joint model of equity and capital; with conversion worth 4bp this is immaterial for BNP, but not for an issuer close to its trigger.

9.4 Extensions

Three extensions follow naturally. A common factor in spreads and capital, estimated on a cross-section of European AT1 issuers, would capture the systemic component of the risk and allow the AT1-specific premium to be decomposed into issuer and sector parts. A model of distributions in which the bank allocates the MDA between dividends and coupons would replace the upper bound on the MDA cost by an estimate. And a joint calibration across issuers, with λ_{res} anchored to each issuer's Tier 2, would turn the residual premium into a relative-value signal.

10 Conclusion

AT1 bonds are claims on bank capital, and modelling them as such makes their three main contingencies, the trigger, the coupon restriction and the call, consistent by construction. Capital alone cannot explain their prices: the model needs a refinancing spread that drives the call and a write-down hazard that rises in the same states. Conditioning on the state paths makes the price an explicit, smooth and convex function of the hazard, so the implied hazard is unique and every comparative static is exact on a given set of paths.

The implied hazard is not a black box. Because Tier 2 faces the same resolution event as AT1, pricing the issuer's Tier 2 on the same state splits the hazard into a part Tier 2 also prices and an AT1-specific premium. For BNP Paribas in September 2026, about four-fifths of the AT1 hazard and two-thirds of the spread are priced in Tier 2 as well, about 50bp is specific to the AT1, and the contract features cost 40–46bp. This split survives a reputational call rule, every attribution that prices the hazard before the contract features, and a term structure of the resolution hazard. What is not robust, and is reported with its range, is the level of capital stress, which depends on how fast the bank rebuilds capital after a shock.

A Further derivations

A.1 Isometry for the compound Poisson integral

Let $\tilde{L}_t = L_t - \lambda_J m_J t$ be the compensated compound Poisson process. For a deterministic square-integrable f, $\mathbb{E}\int_0^t f(u)\,\mathrm{d}\tilde{L}_u = 0$ and

$$\mathbb{E}\Big[\Big(\int_0^t f(u)\,\mathrm{d}\tilde{L}_u\Big)^2\Big] = \lambda_J \mathbb{E}[J^2]\int_0^t f(u)^2\,\mathrm{d}u.$$

Proof. For a step function $f = \sum_k f_k \mathbf{1}_{(u_k,u_{k+1}]}$ the increments of $\tilde{L}$ over disjoint intervals are independent with mean zero and variance $\lambda_J\mathbb{E}[J^2](u_{k+1}-u_k)$, so the second moment is $\lambda_J\mathbb{E}[J^2]\sum_k f_k^2(u_{k+1}-u_k)$. The general case follows by density of step functions in L^2. Writing $\int f\,\mathrm{d}L = \int f\,\mathrm{d}\tilde{L} + \lambda_J m_J \int f\,\mathrm{d}u$ gives the mean and variance used in Proposition 3.2, with $f(u) = e^{-\kappa(t-u)}$.

A.2 Transition of the CET1 process between observation dates

Conditionally on C_s, (3) applied on $[s,t]$ gives

$$C_t = \theta + (C_s-\theta)e^{-\kappa(t-s)} + \sigma\int_s^t e^{-\kappa(t-u)}\,\mathrm{d}W_u - \int_s^t e^{-\kappa(t-u)}\,\mathrm{d}L_u.$$

Without jumps the transition is Gaussian with mean $m = \theta + (C_s-\theta)e^{-\kappa\Delta}$ and variance $\sigma^2(1-e^{-2\kappa\Delta})/(2\kappa)$, $\Delta = t-s$, which is (31). With jumps it is a Gaussian mixture: conditionally on k jumps at times $u_1,\dots,u_k$, the mean is shifted by $-\sum_i J_i e^{-\kappa(t-u_i)}$. The likelihood used in Section 6 ignores this term. Over a quarter the probability of a jump is $1-e^{-0.12/4} \approx 3\%$, so the Gaussian likelihood treats rare large moves as diffusion, which biases $\hat{\sigma}$ upwards slightly and, through the profile likelihood, $\hat{\kappa}$ in an ambiguous direction; the 2023Q1 exclusion removes the one large positive move.

A.3 Profile likelihood

Differentiating (32) with respect to σ^2 gives $\partial\ell/\partial\sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_k (C_{k+1} - m_k)^2/w_k$, which vanishes at (33). Substituting back,

$$\ell_p(\kappa) = \ell\big(\kappa, \hat\sigma(\kappa)\big) = -\frac{n}{2}\Big[1 + \log\big(2\pi\hat\sigma^2(\kappa)\big)\Big] - \frac{1}{2}\sum_k \log w_k(\kappa),$$

a one-dimensional function of κ. The likelihood ratio statistic for a fixed κ_0 is $2[\ell_p(\hat\kappa) - \ell_p(\kappa_0)]$, asymptotically χ^2_1.

A.4 Yield to call and the delta method

Let $y(V)$ solve $V = \sum_j c_j(1 + y/2)^{-2t_j}$ for the bullet cash flows c_j to T_1. By the implicit function theorem

$$\frac{\mathrm{d}y}{\mathrm{d}V} = -\Big[\sum_j c_j t_j (1 + y/2)^{-2t_j - 1}\Big]^{-1} = -\frac{1}{D_{\mathrm{mod}} V},$$

where D_{mod} is the modified duration to call. For a zero-recovery bullet to T_1 with a constant hazard h, the price is the risk-free price discounted at $r + h$, so the continuously compounded yield rises one-for-one with the hazard; in the base attribution the hazard block of 219bp of hazard produces 231bp of semi-annual yield-to-call spread, a ratio of 1.05 that reflects compounding and the state dependence of the driver. The price sensitivity $|V_0'| \approx 0.08$ points per basis point of hazard is larger than a bullet's, because it includes the exposure of extended cash flows beyond T_1. The delta method for the implied hazard, (25), follows from the same implicit function argument applied to $V_0(\lambda) = P$.

A.5 The credit triangle for the CDS reference points

For a CDS with constant hazard h, recovery R and continuous premium s, the protection leg is $(1 - R)\int_0^T h e^{-(r+h)u}\,\mathrm{d}u$ and the premium leg $s\int_0^T e^{-(r+h)u}\,\mathrm{d}u$, so the par spread is $s = h(1 - R)$ and $h = s/(1 - R)$. The CDS rows of Table 4 use this approximation; it ignores the term structure of the hazard, accrued premium on default and the difference between the standard coupon and the par spread, all of which are small at these spread levels.

A.6 Relation of the nested value to the Owen value

For a game v on $\mathcal{N}$ with a partition into unions $\{\mathcal{H}, \mathcal{K}\}$, the Owen value of player $j \in \mathcal{H}$ averages its marginal contribution over all orders in which the members of each union are contiguous, with both orders of the unions equally likely (Owen, 1977). The nested value of Section 5 restricts these orders to those in which $\mathcal{H}$ comes first. It is therefore the Owen value of the game in which the union order is fixed, and it inherits efficiency, symmetry within unions and the null-player property. The full Owen value would average the nested value with the attribution in which the contract block comes first, which values extension on a riskless bond and is excluded for the reason given in Section 8.

B Numerical diagnostics

Monte Carlo error. Table 14 reports, for each AT1 on the valuation date, the path-wise standard deviation of the conditional payoff, the standard error of the price with 20,000 paths, the sensitivity of the price to the hazard and the implied standard error of the hazard from (25).

	6.875% NC33	7.20% NC36	7.45% NC35
Path-wise s.d. of conditional payoff (points)	7.3	7.9	7.5
Standard error of the price (points)	0.051	0.056	0.053
$\lvert\partial V/\partial\lambda\rvert$ (points per bp)	0.077	0.084	0.079
Standard error of λ (bp)	0.67	0.66	0.67

Table 14: Monte Carlo error at the implied hazard, 20,000 paths.

Convergence. Table 15 reports the implied hazards for path counts from 5,000 to 40,000 with two independent seeds, and for a fortnightly instead of a monthly grid. The levels vary by about ±1bp across seeds, in line with the standard errors of Table 14. The cross-bond dispersion is much more stable, at 2.3–2.9bp across all runs, because the three bonds share their random numbers. Halving the time step changes the hazards by at most 1.6bp.

Paths	Seed	6.875% NC33	7.20% NC36	7.45% NC35
5,000	11	2.213	2.193	2.196
5,000	23	2.215	2.183	2.195
10,000	11	2.215	2.191	2.196
10,000	23	2.216	2.195	2.201
20,000	11	2.210	2.186	2.188
20,000	23	2.199	2.176	2.176
40,000	11	2.215	2.184	2.190
40,000	23	2.201	2.176	2.180
20,000, monthly (base)	7	2.211	2.185	2.187
20,000, fortnightly	7	2.227	2.196	2.181

Table 15: Implied hazards (%/yr) by number of paths, seed and time step.

Analytic moments. With the base-case parameters, Proposition 3.2 gives a stationary CET1 mean of 12.25% and a stationary standard deviation of 1.17pp (0.56pp from the diffusion alone); the mean jump is 2.2pp and its 95th percentile 4.4pp. Proposition 3.4 gives a stationary spread standard deviation of 100bp and a stationary probability of a negative spread of 0.3%. The half-lives of shocks are 2.0 years for capital and 1.4 years for the spread.

Memory. The conditional estimator stores, per path, the coupon cash flows and the integrated driver at the coupon dates. Root-finding with closures over these arrays creates reference cycles that the Python garbage collector does not free promptly; forcing a collection after each solve keeps the weekly calibration below 1 GB.

C Parameters

Symbol	Meaning	Value	Source
θ	CET1 target	13%	BNP strategic plan
κ, σ	CET1 mean reversion, volatility	0.35, 0.47pp	base case; MLE 0.165, 0.44
λ_J, m_J, σ_J	stress jumps	0.12/yr, 2.2pp, 0.5	EBA 2025 adverse check
b	MDA floor	5.64%	P1 + P2R (CET1 part)
w_0	combined buffer today	4.79%	published requirement 10.43%
Δw	O-SII step-ups	+0.25pp, 1-Jan-2027 and 2028	French O-SII framework
c_{trig}	trigger	5.125%	term sheets
$a, \bar{s}, \eta, \rho$	spread OU	0.5, 275bp, 100bp, −0.3	sample mean; assumptions
β	capital premium	1% per pp	assumption
δ	reputational tolerance	0 (base), up to 100bp	Section 8
ξ	share price at trigger / today	0.2	assumption
$R_{\text{T2}}, R_{\text{SNP}}$	recovery at resolution	20%, 40%	Markit conventions
λ	AT1 hazard today	2.19% (average)	implied
λ_{res}	hazard priced in Tier 2	1.72%	implied, T2 2035NC34
λ_1, λ_2	two-piece λ_{res}	1.40%, 2.10%	implied, both Tier 2
H, Δ, n	horizon, step, paths	40 years, 1/12, 20,000	numerical

Table 16: Parameters of the base case.

D Data and reproducibility

Data layout. The repository separates raw Capital IQ exports (`data/raw/`, licensed and not redistributed), public reference inputs (`data/reference/`: term sheets, the requirement stack and its sources, the EBA stress test and the FRED curve) and the processed panel built from them (`data/processed/`). A single script, `build_bnp_dataset.py`, produces a daily panel of all six bonds with accrued interest, yield to first call and spread over the FRED curve, together with the CDS, the share price and the CET1 print available on each date, and a quarterly capital table with the requirement and its publication date.

Code map.

File	Content
`processes.py`	CET1 jump-diffusion (1), trigger times
`instrument.py`	MDA schedule (12), stylised term sheet
`market.py`	dated bonds, (7)–(15), pricing (23), implied λ; Tier 2 (27); piecewise hazard
`data.py`	paths, loaders, valuation date
`build_bnp_dataset.py`	raw exports to the processed panel
`calibrate_bnp.py`	CET1 likelihood, weekly implied hazards, cross-section
`capital_structure.py`	Tier 2/SNP/CDS ladder, recovery sensitivity, weekly split
`spread_and_stress.py`	attribution, call analysis, stress, capital delta
`robustness.py`	reputational call, Shapley attributions, term structure
`parameter_uncertainty.py`	bootstrap of (κ, σ), κ scan
`numerics.py`	Monte Carlo errors, convergence, analytic moments

Reproduction. `run_all.sh` rebuilds the panel, runs the test suite (32 tests, including the reduction of the piecewise hazard to the flat one, the round trip of the implied resolution hazard and

the effect of the call tolerance), and regenerates every table and figure in about thirty minutes. Dates, levels and annotations in the figures are read from the data, so newer exports and a new `VALUATION_DATE` update the whole paper.

E Additional figures

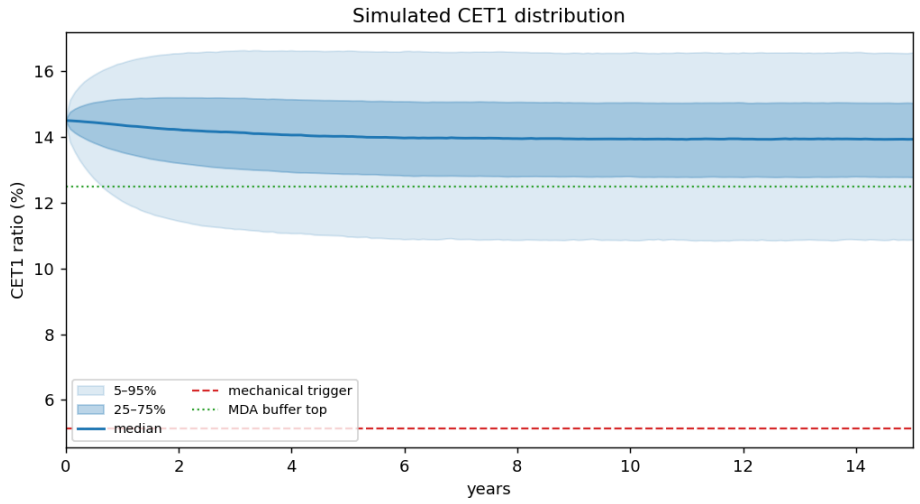


Figure 10: The stylised engine of the repository with illustrative parameters (not the BNP calibration): simulated CET1 distribution over fifteen years, with the top of the MDA buffer and the mechanical trigger.

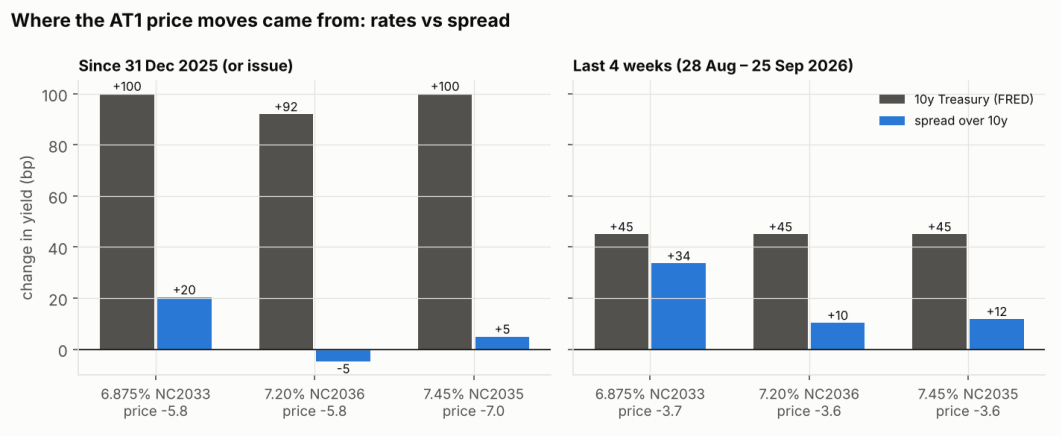


Figure 11: Change in each AT1's yield since 31 December 2025 (or issue) and over the four weeks to 25 September 2026, split into the change in the ten-year Treasury yield and the change in the spread over it.

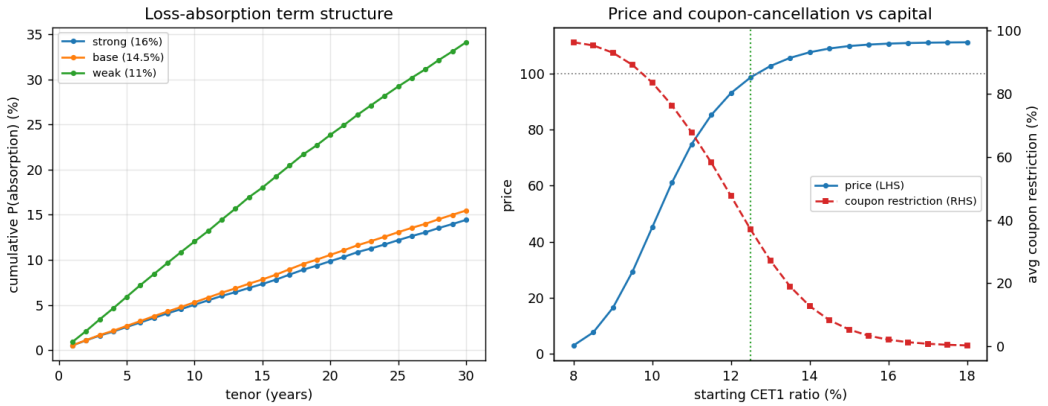


Figure 12: The stylised engine with illustrative parameters: cumulative probability of loss absorption by tenor for a strong, base and weak bank (left), and price and average coupon restriction against the starting CET1 ratio (right).

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